Foundation III — Cosmological Physics

Bounce Signatures, Void Statistics, the Status of the Primordial Spectrum, and Falsifiable Predictions

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Abstract

This document collects the testable, falsifiable cosmological-physics content of the ECF Foundation III programme — the sector developed for peer review. The wormhole mechanism, the Cartan core, the CPT cascade, the full Landau–Ginzburg formalisation of the bounce, and the exploratory programme (baby universes, fertility curve, astrobiology, holarchy) are developed in companion documents (*Foundation III — Strong Gravity*, *Foundation III — Chiral Bounce*, and *Foundation IV*; not yet publicly deposited) and are not repeated here.

Horizon without inflation. The ECF proposes a structural (non-kinematic) alternative to inflation. We establish that the crystallised Kibble–Zurek network forms a genuine chain complex $(\{C_k\}, \partial)$ on the ECKS manifold and *prove* $\partial \circ \partial = 0$ (Proposition F3-1) — a valid algebraic result in its own right. Whether this delivers a globally coherent, causally-connected horizon solution *a priori*, as an earlier version of this Corollary claimed, is not established: that stronger claim required Chiral Walls to be stitched into one object by a network of Torsion String loops, and was withdrawn on 30/08/2026 once Chiral Walls were shown to be independent closed bubbles with no stable multi-way junction. The horizon problem therefore retains an *open* status alongside the monopole problem, documented without occultation (Table 1).

Observational confrontation. The calibrated ECF trajectory $(w_0, w_a) = (-0.904, -0.153)$ is compared to DESI DR1 with full covariance (Mahalanobis distance 2.03σ under a Planck prior). The DES 6-year + DESI DR2 + CMB combination confirms the qualitative ECF direction ($w_0 > -1$, $w_a < 0$) while revealing a tension on the w_a amplitude (PO-F3-DESI). We also state a limit that applies to every (w_0, w_a) model and is not specific to this one: an interacting dark sector can be exactly degenerate with CPL at the background level, so background agreement alone does not identify the underlying mechanism. The discriminant is late-time structure growth, where the ECF/ΛCDM ratio of $f\sigma_8$ rises from 0.870 to 0.959 over $0 < z < 0.8$ (a 9.8% variation) and is not excluded by current redshift-space-distortion data — which is a consistency result, not a preference. The Void Size Function kill-switch against the real DESIVAST DR1 catalogue (29/06/2026) gives $R_{\max} = 31.7$ Mpc/h — the ECF threshold at $\xi_{\rm KZ} = 60$ comoving Mpc (40.4 Mpc/h at $h_{\rm fid} = 0.6732$) remains out of reach of the BGS sample ($z < 0.24$); the method is validated and the definitive test deferred to Euclid DR1-Complete (mid-2027, PO-F3-VSF).

Primordial spectra. The non-Gaussianity of the KZ network ($f_{\rm NL}^{\rm KZ}$) and the primordial tensor spectrum from the stiff phase are developed as physical templates for a future perturbative solver, not as closed predictions. The scalar spectral index is *not* predicted by the framework: the large-scale spectrum is taken as inherited (PO-F3-NS), while the stiff era generates a blue component ($\mathcal{P}_{\mathcal{R}} \propto k^3$) whose amplitude bounds the stiff duration from below. The *form* of that bound is falsifiable; its numerical threshold is not yet calibrated, since it depends on an $a(T)$ relation the framework does not determine (NEFOLD-1).

Open problems documented without occultation. PO-F3-ISO (isotropy), PO-F3-NS (spectral index, principal open front), PO-F3-XI ($\xi_{\rm KZ}$ not derived *ab initio*), PO-F3-DESI (w_a tension), PO-F3-VSF (kill-switch out of BGS reach). Strong-gravity and chiral-

sector open problems (PO-F3-C6, PO-F3-B3, PO-F3-IC, PO-F3-CHIRAL) are tracked in the companion *Foundation III — Strong Gravity* document.

Keywords: Einstein–Cartan cosmology · non-singular bounce · Kibble–Zurek network · chain complex · horizon problem without inflation · scalar spectral index · primordial spectrum (inherited) · stiff phase · primordial tensor spectrum · non-Gaussianity · Void Size Function · DESI BAO · dark energy · BBN constraint · CMB isotropy · gravitational-wave background · falsifiability

Mots-clés : cosmologie d'Einstein–Cartan · rebond non singulier · réseau de Kibble–Zurek · complexe de chaînes · problème de l'horizon sans inflation · indice spectral scalaire · spectre primordial (hérité) · phase *stiff* · spectre tensoriel primordial · non-gaussianité · fonction de taille des vides · BAO DESI · énergie noire · contrainte BBN · isotropie du CMB · fond d'ondes gravitationnelles · falsifiabilité

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Scope and boundaries of this document (Charter, SPLIT-1)

Covers: testable, falsifiable cosmological physics — BBN constraint; DESI DR1/DR2 comparison with full covariance; status of the primordial spectrum (large-scale tilt inherited, not predicted; the stiff era's blue component bounds $N_{\rm stiff}$); VSF kill-switch vs DESIVAST (a robustness test, not a blind prediction); PO-F3-ISO (isotropy); non-Gaussianities $f_{\rm NL}^{\rm KZ}$; primordial tensor spectrum and stiff-phase amplification; structured kill-switch table; positioning vs Hartle–Hawking–Hertog.
Explicitly excludes (see companion documents): the detailed Landau–Ginzburg potential and parity-breaking mechanism (*Foundation III — Chiral Bounce*); the wormhole mechanism, baby-universe fusion, and Cartan core (*Foundation III — Strong Gravity*); and the exploratory programme, holarchy genesis and cosmological information (*Foundation IV*).
Evolution rule: any new falsifiable cosmological prediction belongs here; any speculative extension belongs in a companion document, even when related.

1 Introduction

Foundation III is the third instalment of the Einstein-Cartan Framework (ECF) corpus, following PIT [6] (exact spatial flatness as a topological postulate) and Foundation I–II [7, 8] (ECKS bounce, KZ topological network, bounce fossils). This manuscript collects the testable, falsifiable cosmological-physics content of the Foundation III programme — horizon coherence without inflation, the WKB status of the primordial spectrum, and the observational confrontation with DESI and the Void Size Function. The strong-gravity sector (the Cartan core, the Einstein-Rosen bridge, the CPT cascade) and the exploratory programme (biological implications, baby universes) are developed in the companion documents *Foundation III — Strong Gravity*, *Foundation III — Chiral Bounce*, and *Foundation IV* (not yet publicly deposited), honouring the explicit commitments made to Foundation III in the earlier papers, and are not repeated here.

The main results reported here are: (i) the formal derivation $\partial \circ \partial = 0$ on the ECKS chain complex (Proposition F3-1, §2.1); (ii) the WKB analysis of the ECKS Wheeler-DeWitt potential and isotropy selection under the Vilenkin prescription (§3.3); (iii) the DESI DR1 comparison with full covariance ($d_{\rm Maha} = 2.03\sigma$, Planck prior; §5.2) and the updated DES 6-year + DESI DR2 combination (3.0σ vs ΛCDM, §5.2); (iv) the preliminary Void Size Function kill-switch test against DESIVAST DR1 (§3.2).

The five central open problems (PO-F3-ISO, PO-F3-NS, PO-F3-XI, PO-F3-DESI, PO-F3-VSF) are documented without occultation throughout the text. The chiral-sector problem (PO-F3-CHIRAL) is touched upon here only in the context of the observational homochirality discussion (Table 2); its full treatment is tracked in the companion *Foundation III — Strong Gravity*. Three further threads complete the corpus: (vi) the resolution of the information paradox via topological encryption through the Weyssenhoff fluid (see *Foundation IV*); (vii) transfer scaling laws and the spectral contrast — *where what the ECF predicts is the blue enhancement of the energy spectrum transferred through the stiff phase, to be distinguished from the primordial tilt* $n_T^{\rm prim}$ *itself* — $n_T > 0$ (ECF) vs $n_T < 0$ (inflation) as a tensor kill-switch (see *Foundation III — Strong Gravity*); (viii) the ECF Fertility Curve and the multiverse Goldilocks Zone delimiting the conditions for the emergence of complex life (see *Foundation IV*).

2 Structural Alternative to Inflation

2.1 Beyond the Standard Model and Inflation

The horizon problem: precise formulation

Scalar-field inflation solves three problems simultaneously: the horizon, flatness, and magnetic monopoles. The horizon problem can be stated as follows: at recombination ($z \sim 1100$), two

CMB regions separated by more than $\sim 2^\circ$ were not in causal contact in a standard Friedmann-Robertson-Walker universe. The alternative mechanism must therefore produce the equivalent of $N \gtrsim 60$ e-folds of causal connection.

In the *stiff* phase ($w = 1$, $a \propto t^{1/3}$), the comoving Hubble radius *grows*:

$$\frac{d}{dt}\left(\frac{1}{aH}\right) > 0 \quad (\textit{stiff}\text{ phase}) \qquad \text{vs} \qquad \frac{d}{dt}\left(\frac{1}{aH}\right) < 0 \quad (\text{inflation}), \tag{1}$$

which means the *stiff* phase does *not* solve the horizon problem through the same kinematic mechanism as inflation. *The ECF proposes a fundamentally different resolution: structural rather than dynamical.* This is not a matter of taste. Izaurieta, Lepe and Quinzacara [33] compute the budget of the post-bounce accelerated window for a Weyssenhoff spin fluid with barotropic source $w \in [0, 1]$ and find $N_e = \ln[4/(1+3w)]/[3(1-w)] \leq \frac{1}{3}\ln 4 \simeq 0.46$ e-folds — two orders of magnitude short of inflationary needs, for any barotropic source and with no tuning available; positive spatial curvature only shrinks the window further. The kinematic route is therefore closed, and the present framework does not take it.

Resolution via topological network: the constraint $\partial \circ \partial = 0$

At the electroweak crystallisation that follows the ECF bounce ($T_{\rm EW} \sim 159\,\text{GeV}$; the bounce itself occurs near the Cartan density, $T_{bb} \sim 10^{19}\,\text{GeV}$, Eq. (32)), the Kibble–Zurek mechanism crystallises a network of topological defects (see Foundation II, § *Topological Zoology* [8]):

- **0D Micro-Knots** (π_2) are point defects, the sector retained in every vacuum-manifold assignment considered (the group-theoretic derivation of $\mathcal{M}$ is open, PO-F2-8).
- **2D Chiral Walls** (π_0) are independent closed surfaces, *conditional* on the effective potential, on the vacuum manifold (not derived, PO-F2-8) and on the symmetry not being restored (I69, 30/08/2026: no stable multi-way junctions for a $\mathbb{Z}_2$ wall sector, hence no connected network of the kind pictured below).
- The **3D Torsion Fluid** fills the space between them, established.
- A candidate **1D Torsion String** sector (π_1) was investigated and is retained as an explicitly open problem (Foundation II, PO-F2-8): S^μ is real and carries no internal phase, so the Lagrangian does not currently supply the complex order parameter such a defect requires.

This network satisfies the fundamental algebraic constraint:

$$\partial \circ \partial = 0. \tag{2}$$

The boundary of a boundary is zero — a general algebraic identity, true of any chain complex regardless of the physical system it describes. **It does not, by itself, deliver a globally coherent, causally connected network**: that stronger claim required every Chiral Wall to be bounded by loops of Torsion Strings, stitching all bubbles into one object, and was withdrawn on 30/08/2026 once Chiral Walls were shown to be independent closed surfaces with no stable multi-way junction (I69, above). The formal development below is retained because the underlying algebra is correct and instructive; a correction box further down makes explicit, at the point where the physical over-reach occurs, exactly what does and does not survive.

Formal proof: $\partial \circ \partial = 0$ via the ECKS chain complex

Construction of the chain complex. Let $\mathcal{M}_{\rm ECKS}$ be the Einstein–Cartan–Kibble–Sciama manifold at electroweak crystallisation ($T \sim T_{\rm EW}$), after the bounce. The four types of topological defects in the KZ network correspond to four groups of singular chains with coefficients in $\mathbb{Z}$:

$$0 \longrightarrow C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \longrightarrow 0, \tag{3}$$

where the chain spaces are physically identified as follows. These identifications inherit an open question from Foundation II: the vacuum manifold carrying π_0, π_1 and π_2 simultaneously has not been derived (PO-F2-8), and the assignment $\mathcal{M} \simeq S^2$ quoted there would in fact admit only the 0D sector. The chain complex below is therefore built on assignments that are physically motivated but not yet grounded in a group-theoretic derivation.

- C_0: formal linear combinations of **Micro-Knots** (0D defects, π_2);
- C_1: formal linear combinations of oriented **Torsion Strings** (1D defects, π_1);
- C_2: formal linear combinations of oriented **Chiral Walls** (2D defects, π_0);
- C_3: formal linear combinations of Torsion Fluid **domains** (3D volumes).

The boundary operators $\partial_k : C_k \to C_{k-1}$ act geometrically as:

$$\partial_1(\sigma_{\text{str}}) = \sigma_{\text{str}}(1) - \sigma_{\text{str}}(0) \quad \text{(endpoint minus origin of the Torsion String)}, \tag{4}$$
$$\partial_2(\Sigma_{\text{wall}}) = \partial(\Sigma_{\text{wall}}) \quad \text{(boundary loop of the Chiral Wall)}, \tag{5}$$
$$\partial_3(V_{\text{fluid}}) = \partial(V_{\text{fluid}}) \quad \text{(boundary surface of the 3D domain)}. \tag{6}$$

Fundamental property $\partial \circ \partial = 0$.

Proposition F3-1 (coherence of the ECKS network)

For all $k \in \{1, 2, 3\}$ and every chain $c \in C_k$:

$$\partial_{k-1} \circ \partial_k \,(c) = 0.$$

Proof. The property follows from the geometric definition of the iterated boundary at each degree.

Degree 1 ($\partial_0 \circ \partial_1 = 0$). For a Torsion String $\sigma : [0, 1] \to \mathcal{M}_{\text{ECKS}}$,

$$\partial_1(\sigma) = \sigma(1) - \sigma(0).$$

Applying ∂_0 (boundary of a 0-simplex, identically zero by convention):

$$\partial_0 \circ \partial_1(\sigma) = \partial_0(\sigma(1)) - \partial_0(\sigma(0)) = 0 - 0 = 0. \quad \checkmark$$

Degree 2 ($\partial_1 \circ \partial_2 = 0$). For a Chiral Wall $\Sigma \subset \mathcal{M}_{\text{ECKS}}$ with boundary $\partial_2(\Sigma) = \partial\Sigma$ (a closed curve),

$$\partial_1 \circ \partial_2(\Sigma) = \partial(\partial\Sigma) = 0,$$

since the boundary of a closed curve (a loop formed by Torsion Strings) is the algebraic sum of its vertices counted with oriented multiplicity, which cancel pairwise. $\checkmark$

Degree 3 ($\partial_2 \circ \partial_3 = 0$). For a Torsion Fluid domain $V \subset \mathcal{M}_{\text{ECKS}}$, $\partial_3(V) = \partial V$ is a closed surface (no boundary). Hence $\partial_2(\partial V) = \partial(\partial V) = 0$. $\checkmark$

In every case, $\partial_{k-1} \circ \partial_k = 0$ follows from the fact that *the boundary of a boundary is a manifold without boundary.* ■

Link with de Rham cohomology on $\mathcal{M}_{\text{ECKS}}$. The de Rham theorem establishes a natural isomorphism between singular homology with real coefficients and the cohomology of differential forms:

$$H_k(\mathcal{M}_{\text{ECKS}};\mathbb{R}) \;\cong\; H^k_{\text{dR}}(\mathcal{M}_{\text{ECKS}}), \tag{7}$$

where $H^k_{\text{dR}} = \ker d_k / \operatorname{im} d_{k-1}$ with d the exterior derivative ($d \circ d = 0$ being the dual formulation of $\partial \circ \partial = 0$ via the generalised Stokes formula).

Concretely, every topological defect in the KZ network corresponds to a non-trivial cohomology class on $\mathcal{M}_{\text{ECKS}}$:

- $H^0_{\text{dR}} \cong \mathbb{Z}^{N_0}$, where N_0 is the number of connected components of the Torsion Fluid (Micro-Knots as generators);
- H^1_{dR} captures the non-contractible loops of Torsion Strings;
- H^2_{dR} captures the closed surfaces of Chiral Walls.

The ECKS torsion constraint requires the Betti numbers $b_k = \dim H^k_{\text{dR}}$ to be finite and constrained by the ECF geometry at the bounce — an explicit derivation is deferred to the a dedicated Einstein–Cartan perturbation solver (Foundation IV, §*ECKS Topology*).

Physical consequence: global coherence without inflation. The property $\partial \circ \partial = 0$ guarantees that the KZ network is a *cycle* in the homological sense: $\partial(\text{network}) = 0$. This means there is no "free end" anywhere in the network — every Torsion String terminates on Micro-Knots, every Chiral Wall is bounded by loops of Torsion Strings — across *the entire manifold* $\mathcal{M}_{\text{ECKS}}$, regardless of the comoving separation between regions.

Consequence (Corollary F3-1): the global topological coherence of the KZ network is not a consequence of thermal equilibration, but an *a priori* property of the chain complex $(\{C_k\}, \partial)$ on $\mathcal{M}_{\text{ECKS}}$. It functionally replaces the $N \gtrsim 60$ e-folds of inflation for resolving the horizon problem, *without* requiring a scalar inflaton field.

Correction of record (30/08/2026) — Corollary F3-1 withdrawn as an established result

Corollary F3-1 as stated above is **withdrawn as an achieved result**. Its premise — "every Chiral Wall is bounded by loops of Torsion Strings" — has since been shown false (Foundation II, I69, 30/08/2026): 2D Chiral Walls are independent closed surfaces ($\mathbb{Z}_2$ topology, no stable multi-way junctions), each already without boundary on its own, *not* open sheets stitched together by a 1D string network. Without that stitching, the KZ network is not one globally connected object: it is a large number of causally disconnected, individually closed bubbles. $\partial \circ \partial = 0$ for the wall itself remains trivially true (a closed surface has no boundary, by definition), but this no longer supplies the *global* coherence the Corollary claimed — there is no mechanism here connecting one bubble's interior to another's, which is precisely the function the horizon problem requires.

What survives. The purely 2D/3D content ($\partial_2 \circ \partial_3 = 0$ for a closed wall bounding a fluid domain) is unaffected and remains a valid, if now unremarkable, geometric statement.

What does not. The claim that this replaces inflation's horizon-problem solution is withdrawn. Restoring it would require *two* results, neither established: (i) a demonstration that the 1D Torsion String sector exists with genuine π_1 winding — partial progress is reported in Foundation II, PO-F2-8 (30/08/2026): the stated collinearity obstruction between the axial vector S_μ and its trace-vector companion is resolved near a defect by an explicit, verified geometric mechanism, but no azimuthal winding has yet been exhibited; and (ii), independently and not yet attempted, a demonstration that such strings, if they exist, actually reconnect the wall network into one globally coherent object rather than merely decorating each bubble's boundary individually. Table 1 below is corrected accordingly: the Horizon row is downgraded from *Covered* to *Open*, and the Monopoles row's candidate track is noted as sharing the same dependency.

Range of validity: this proof assumes the topology of $\mathcal{M}_{\text{ECKS}}$ is that of a smooth, compact, orientable manifold at the time of the bounce. Regularisation of the bounce singularity by torsion pressure (Foundation I, § *Torsion Bounce* [7]) is required for this framework to apply. The extension to the non-orientable case (CPT breaking) is an open problem (Foundation IV).

Comparative table: cosmological problems, inflation vs ECF

Scalar-field inflation historically solves five problems. Table 1 summarises, for each, *how* the ECF addresses it — and honestly flags the cases where coverage is partial or unestablished. The central message is that the ECF secures the *functions* of inflation through distinct mechanisms (structural rather than kinematic), without an inflaton field, for three of the five problems; **two — horizon and monopoles — retain an open status (updated 30/08/2026)**, sharing a common, withdrawn premise (see correction box above Table 1).

On the monopole problem (rigorous analysis, epistemic honesty). This problem deserves careful examination, since the ECF lacks inflation's dilution mechanism, and two intuitive arguments prove *incorrect* on analysis:

- *Erroneous argument #1 — "the bounce is electroweak, below the GUT scale".* Two temperatures must be distinguished: the KZ-network crystallisation scale ($T_{\text{EW}} \sim 159\,\text{GeV}$) and the temperature of the bounce *itself* ($T_{\text{bb}} \approx 0.8\,T_{\text{Pl}} \sim 10^{19}\,\text{GeV}$, PO-F3-4; this is the route of Popławski, $\varepsilon_{\text{bb}} = 15\,\varepsilon_{\text{Pl}}$, distinct from the Cartan threshold $\rho_{\text{spin}} = \rho_{\text{rad}}$ at $0.49\,E_{\text{Pl}}$ computed in `Scripts_F1/script_07_cartan_threshold.py`, the two agreeing to 40% *(clarified 15/10/2026; the ratio $T_{\text{bb}}/T_{\text{EW}}$ below is corrected from* 6.6 *to* 6.1×10^{16}*, $N_{\text{stiff}} \approx 39$ unchanged)*). The bounce is therefore *trans-GUT*: while cooling, the child universe *crosses* the GUT scale ($\sim 10^{16}\,\text{GeV}$), making possible a restoration and subsequent re-breaking of GUT symmetry — and hence monopole formation via the Kibble mechanism. The claim "we never cross the GUT scale" is *false*.
- *Erroneous argument #2 — "the stiff phase dilutes the monopoles".* The stiff-phase density decays as a^{-6}, *faster* than non-relativistic monopoles (a^{-3}). Unlike inflation (where ρ_{inflaton} stays nearly constant and dilutes everything else), the stiff phase sees monopoles *gain* relative importance. It therefore *does not help* — the opposite of inflation on this point.

Correct status: open challenge, unresolved. If grand unification exists *and* is restored at the bounce, the ECF faces the monopole problem without the inflationary mechanism. One point is nonetheless established: any monopoles from the *parent* are not transmitted to the child (only the seed E_{rot} crosses over, PO-F3-1; massive defects remain with the irreducible mass), so any monopole population would arise from *local* formation in the child. Two unproven resolution tracks: *(i) absence of grand unification* — proton decay has never been observed (Super-Kamiokande), but this argument is conditional and not specific to the ECF; *(ii) the constraint* $\partial \circ \partial = 0$ (Proposition F3-1) was proposed as a track that could forbid isolated topological charges; **this track shared the premise of Corollary F3-1, withdrawn on 30/08/2026** (a connected network without free ends), and currently has no independent support (Table 1); it would not in any case reach GUT monopoles, which live in the internal gauge bundle, not in the spatial KZ network. **The status is therefore "open problem, potentially serious"**, to be addressed in Foundation IV — explicitly *not* an achieved ECF success.

Investigation brief — MONO-1 (15/07/2026)

The exact crux of the problem. The ECKS chain complex (Eq. (3)) lives in *physical spacetime*: it is a constraint on the *spatial* defects of the KZ network. GUT monopoles, by contrast, live in the *internal gauge bundle*: they are classified by $\pi_2(G/H)$, the homotopy group of the gauge theory's vacuum manifold — an abstract space of symmetry-breaking directions, not physical space. **These are two *a priori* distinct mathematical spaces**, and $\partial \circ \partial = 0$ says *nothing* directly about

Problem	**Inflation solution**	**ECF treatment**	**Status**
Horizon	Kinematic contraction of the comoving Hubble radius ($N \gtrsim 60$ e-folds)	*A priori* topological coherence of the KZ network ($\partial \circ \partial = 0$, Prop. F3-1) — *structural*, not kinematic; **premise withdrawn 30/08/2026** (walls are independent closed bubbles, not stitched by a connected string network; see correction box above)	**Open** (previously claimed Covered)
Flatness	Exponential dilution of curvature ($\Omega \to 1$)	Zero-total-energy growth ($E_{\rm mat} + E_{\rm grav} \simeq 0$) — flat by construction	Covered (dynamical derivation to be formalised)
n_s spectrum	Inflaton slow-roll	Not predicted: large-scale spectrum inherited; stiff era adds a blue component ($\mathcal{P}_\mathcal{R} \propto k^3$)	**Not covered** — taken as an initial condition (PO-F3-NS)
Primordial fluctuations	Quantum fluctuations of the inflaton	KZ network of topological defects (3D Poisson), $f_{\rm NL} > 0$ conditional on the Poisson$\to \zeta$ bridge; CMB projection not established (§3.1)	**Conditional** (distinct mechanism; not shown to source the observed spectrum, PO-F3-NS)
Magnetic monopoles	Exponential dilution below observational threshold	The bounce is *trans-GUT* ($T_{\rm bb} \sim 10^{19}$ GeV): the child crosses the GUT scale while cooling. The stiff phase provides no dilution (it decays as a^{-6}, faster than monopoles). Candidate track: the constraint $\partial \circ \partial = 0$ could forbid isolated topological charges — **this candidate track shares the same withdrawn premise as the Horizon row above (30/08/2026) and currently has no independent support**	**Open (challenge, no candidate track pending 30/08/2026)**

Table 1: Comparison of how scalar-field inflation and the ECF cover the classic cosmological problems. **Corrected 30/08/2026**: the ECF covers *three* of the five problems through *structural* mechanisms distinct from inflation, without an inflaton (Flatness, n_s is explicitly not covered, Primordial fluctuations). The Horizon row, previously claimed covered via Prop. F3-1, is corrected to Open: the premise that Chiral Walls are stitched into one connected object by a Torsion String network does not hold, since the walls are independent closed bubbles (Foundation II, I69). Monopoles remain an *open challenge*: since the bounce is trans-GUT ($T_{\rm bb} \sim 10^{19}$ GeV), the child universe crosses the Grand Unification scale while cooling, and the stiff phase provides no dilution (it decays faster than monopoles); its previously proposed

$\pi_2(G/H)$. This is the missing "bridge".
The mathematical question to settle. Does a natural map exist between the spatial KZ network and the internal gauge bundle, such that the constraint $\partial \circ \partial = 0$ on the former imposes a constraint on the latter? Three sub-questions, in increasing order of difficulty:

1. Does a torsion–gauge coupling exist in the full ECKS Lagrangian (beyond the coupling to fermionic spin via Hehl–Datta)? Without it, no physical mechanism links the two spaces, and the track collapses immediately.
2. If so, what is its form? Would a minimal coupling ($S^a \partial_a \phi$, ϕ a GUT field) suffice to transport the network's topological constraint to the gauge bundle?
3. Would the transported constraint be *selective*? It would need to specifically forbid monopoles (and not, say, cosmic strings or GUT domain walls), or it would merely relocate the problem.

Short, decisive preliminary checks (before any torsion–gauge coupling calculation, which is heavy) :

- **[A]** Recompute ξ_{KZ} at the GUT scale ($T \sim 10^{16}\,\mathrm{GeV}$) at the time of the bounce — *not* the late-time value ($\sim 60\,\mathrm{Mpc}$) used elsewhere in the corpus, which concerns a distinct regime.
- **[B]** Estimate Ω_{mono} within the ECF framework (bounce temperature, $\xi_{\mathrm{KZ}}(T_{\mathrm{GUT}})$) and compare with the standard Preskill/Zel'dovich–Khlopov result ($\Omega_{\mathrm{mono}} h^2 \sim 10^{11}$ without inflation): does the ECF framework already change the order of magnitude *before* invoking $\partial \circ \partial = 0$?
- **[C]** Search the ECKS Lagrangian for a torsion–gauge coupling (sub-question 1 *above*).
- **[D]** If [C] is positive: compute the transport of the constraint (sub-questions 2–3) — a heavy calculation, closing MONO-1 or not.

Recommended order: [A] and [B] are direct calculations, independent of the $\partial \circ \partial = 0$ track, and already tell us whether the problem is as severe as the standard case (10 orders of magnitude) or already mitigated within the ECF framework specifically — which then guides investment in [C] and [D].

[A]+[B] — first results (15/07/2026). Both preliminary checks have been carried out.

[A]. The Kibble–Zurek correlation length at the GUT scale ($T_{\mathrm{GUT}} \sim 10^{16}\,\mathrm{GeV}$, not to be confused with the *late-time* value $\sim 60\,\mathrm{Mpc}$ used elsewhere in this document for a distinct regime) follows from the mean-field scaling law already established (item LG-6, $\nu = 1/2$, $z = 2$):

$$\xi_{\mathrm{KZ}}(T_{\mathrm{GUT}}) \;\sim\; \xi_0 \left(\frac{\tau_Q}{\tau_0}\right)^{1/4} \;\approx\; 5.8 \times 10^{-32}\,\mathrm{m} \;\approx\; 3\,\xi_0, \tag{8}$$

where $\xi_0 \sim \hbar c / T_{\mathrm{GUT}}$ and $\tau_Q \sim 1/H(T_{\mathrm{GUT}})$ (standard radiative expansion rate at this scale). This result, of order a few ξ_0, matches the order of magnitude expected from the Kibble mechanism.

[B]. Applying the standard formula $n_{\mathrm{mono}} \sim p/\xi_{\mathrm{KZ}}^3$ ($p \sim 0.3$) and the usual radiative dilution to today, one obtains

$$\Omega_{\mathrm{mono}} \;\sim\; 10^{21}\text{–}10^{24}, \tag{9}$$

for $M_{\mathrm{mono}} = 10^{16}$–$10^{18}\,\mathrm{GeV}$, with or without the entropy factor $g_{*s,0}/g_{*s}(T_{\mathrm{GUT}})$ (`estimate_omega_mono_F3.py`) *(corrected 15/10/2026: earlier versions gave 10^{24} with no producing script; that value corresponds to the upper end, $M_{\mathrm{mono}} \sim 10^{18}$ GeV without the entropy factor)*, i.e. a disaster *at least as severe* as Preskill's historical estimate (1979) without inflation ($\Omega_{\mathrm{mono}} h^2 \sim 10^{11}$): the problem is **not** mitigated by the framework's generic ingredients alone. *Status of this figure*: order-of-magnitude, sensitive to the monopole mass and the Kibble fraction p — not a precision result, but sufficient to establish that no automatic mitigation operates.

Unexpected finding — a link with NEFOLD-1. While searching for whether an ingredient *specific to the ECF* could mitigate Ω_{mono} — in particular the stiff phase interposed between the bounce and the usual radiative epoch, which would modify the $a(T)$ relation used for dilution —

the calculation runs into an unknown already identified elsewhere: **the $a(T)$ relation during the stiff phase is not determined in the current state of the corpus**, exactly the obstacle blocking *NEFOLD-1* (a characterised, unresolved item — see the discussion of n_s *ab initio*). **MONO-1 and NEFOLD-1 thus share a common unknown**: the precise thermodynamics of the stiff phase. Progress on either could therefore unblock the other — and, conversely, as long as this relation remains undetermined, neither can be closed independently.

KZ correlation length and cosmological dilution

The Kibble–Zurek correlation length at the bounce is estimated from the mean geometry of the correlation cell at $T_{\rm EW}$: the geometric mean of the Compton length $\hbar c/(k_B T_{\rm EW})$ and the horizon length $c/H_{\rm EW}$ at that temperature,

$$\xi_{\rm KZ}^{\rm bounce} \sim \sqrt{\frac{\hbar c^2}{k_B T_{\rm EW} \cdot H_{\rm EW}}} \sim 8.3 \times 10^{-11}\ {\rm m}, \tag{10}$$

with $H_{\rm EW}$ from the standard radiation-era Friedmann equation, $H = 1.66\sqrt{g_*}\,T^2/M_{\rm Pl}$ at $T_{\rm EW} = 159\,{\rm GeV}$ with $g_* = 106.75$ (the same relation that fixes $M_{\rm MK} = c^3/2GH_{\rm EW}$ in Foundation II), giving $H_{\rm EW} \approx 5.40 \times 10^{10}\,{\rm s}^{-1}$. **Correction of record (05/09/2026):** the formula above was dimensionally inconsistent as originally written (a missing power of c), and the value previously quoted (1.2×10^{-14} m) could not be reproduced from any $H_{\rm EW}$ convention in this corpus, nor recalled by the author. Both defects are corrected together here.

The redshift from the EW epoch to today is:

$$1 + z_{\rm EW} \approx \frac{T_{\rm EW}}{T_0}\left(\frac{g_{*s}(T_{\rm EW})}{g_{*s,0}}\right)^{1/3} = \frac{159\,{\rm GeV}}{2.35 \times 10^{-4}\,{\rm eV}}\left(\frac{106.75}{3.91}\right)^{1/3} \approx 2.0 \times 10^{15}, \tag{11}$$

the value used in Foundation II. *(Corrected 15/10/2026: earlier versions omitted the entropy factor $(g_{*s}/g_{*s,0})^{1/3} \simeq 3.0$ and gave 6.8×10^{14}; the diluted length below rises from 56 to ∼ 170 km and the gap falls from ∼ 19.5 to ∼ 19.0 orders of magnitude, without changing the conclusion.)*

The diluted comoving correlation length is then $\xi_{\rm KZ}^{\rm bounce} \times (1 + z_{\rm EW}) \sim 1.7 \times 10^5$ m ≈ 170 km — **several orders of magnitude below the calibrated value $\xi_{\rm KZ} \sim 50$–80 Mpc**.

PO-F3-XI — microscopic origin of $\xi_{\rm KZ}$

Direct KZ dilution from $T_{\rm EW}$ does *not* produce $\xi_{\rm KZ} = 60$ Mpc. The gap is ∼ 19 **orders of magnitude** (not "several", as an earlier version of this note stated, nor the previously miscalculated "∼ 24"): $\xi_{\rm KZ}^{\rm bounce} \times (1 + z_{\rm EW}) \sim 1.7 \times 10^5$ m against a calibrated 60 Mpc $\sim 1.9 \times 10^{24}$ m. The value $\xi_{\rm KZ} \sim 50$–80 Mpc is an **observational calibration** against the Void Size Function (Foundation II, §VSF [8]); it is not an *ab initio* prediction from microscopic parameters.

Hypothesis (i) — wrong transition scale — now ***quantitatively excluded*** (16/07/2026): moving back from $T_{\rm EW}$ (159 GeV) to $T_{\rm QCD}$ (200 MeV), or even to the Great Annihilation ($T \sim 1$ MeV), regains only 2–3 orders of magnitude out of the ∼ 19 required (the comoving dilution factor grows only as $T^{-1/2}$). No Standard Model transition can close this gap by a simple change of scale.

Hypothesis (ii) — network frozen at the causal-horizon scale (16/07/2026, **recomputed on the ECF background 26/08/2026**): if the topological network remains locked to the comoving causal horizon (*scaling* behaviour, standard for defect networks) until a late freeze-out, then $\xi_{\rm KZ}$ fixes that freeze-out redshift.

Correction of record. The figures previously quoted here ($z \approx 5670$ for the instantaneous Hubble radius, $z \approx 6430$ for the particle horizon, and a factor 1.17 against the acoustic window $z \approx 7500$) were obtained on a ΛCDM background. That is not the background of this framework: the stiff spin fluid contributes a $(1+z)^6$ term which raises H at early times and therefore *shortens* the comoving horizon, by a factor $\simeq 0.68$ near $z \approx 7500$. Recomputing on the ECF background itself gives, for $\xi_{\rm KZ} = 60$ Mpc, $z \approx 5976$ (Hubble radius) and $z \approx 5145$ (particle horizon). *Second correction, 05/09/2026:* these were previously compared against $z \approx 7500$, giving factors 1.25 and 1.46. That comparison used the

wrong anchor: the spin sector is normalised at $z_{\rm ref} = 5600$ in `script_01_solve_sound_horizon.py`, the value that gives $r_s = 135.65\,$Mpc, not at 7500 (see I82). Against $z_{\rm ref} = 5600$ the factors are 0.94 (Hubble radius) and 1.09 (particle horizon) — both within 9%.

Two consequences follow. First, the claim that "the agreement tightens in moving from the Hubble radius to the full particle horizon" does not hold, though only marginally: against $z_{\rm ref} = 5600$ the offset grows from 6.3% (Hubble radius) to 8.8% (particle horizon), so the fuller treatment moves the freeze-out slightly *away* from the normalisation redshift rather than towards it. Second — and contrary to what the 30/08 version of this note concluded on the wrong anchor — the numerical proximity is *tighter* than was then stated, not weaker: under 10% rather than the 25–46% obtained against $z \approx 7500$. This does not make it evidence: a sub-10% coincidence between two quantities of the same order, one of which ($\xi_{\rm KZ}$) is itself calibrated rather than derived, is suggestive at best. What survives is the overlap with HeIII→HeII recombination ($z \sim 5000$–8000): $z \approx 5145$ falls inside that range, though near its lower edge rather than in its middle.

We record this as a correction rather than a result. The hypothesis remains open and is not strengthened by the horizon argument in the way previously written; PO-F3-XI is unchanged in status. *Reproduced by:* `Scripts_F3/check_xi_kz_horizon.py`, which imports the calibrated background from `script_01_solve_sound_horizon.py` so that the two cannot drift apart.
This is no longer an isolated numerical coincidence: the topological network's freeze-out would date to an epoch already known to mark a state change of the primordial plasma.

Horizon tracking is derived, not assumed (17/07/2026). The self-accelerating mechanism $\ddot{z} = (8\pi G/3)(\sigma_{2D}/ct)\,z$ described in Foundation II (PO-F2-3, $\sigma_{2D} \sim 10^{49}\,$kg/m^2) has a characteristic time $\tau(t) = \sqrt{3ct/(8\pi G\sigma_{2D})}$, from $\sim 7 \times 10^{-22}\,$s at the EW epoch to $\sim 2 \times 10^{-16}\,$s at the Great Annihilation — 10 to 16 orders of magnitude below the Hubble time at *any* relevant cosmological epoch (EW, QCD, Great Annihilation). *(Corrected 15/10/2026: earlier versions wrote* $\tau = \sqrt{3/(8\pi G\sigma_{2D})} \approx 1.3 \times 10^{-20}\,s$ *and "9 to 20 orders", dropping the factor ct that converts the surface density into a volume density — the defect corrected in Foundation II on 15/09/2026.)* The wall physically cannot grow at its "natural" rate (which would violate causality): it is therefore constrained, from its formation, to track the causal horizon. **Horizon tracking (hypothesis ii) is thus a derived consequence of the wall tension already present in F2, not an ad hoc assumption**. This same calculation shows, however, that τ remains microscopic well beyond $z \approx 5145$: this mechanism alone therefore does not explain *why* tracking would stop (freeze) precisely at this epoch rather than another — this requires the chiral annihilation dynamics (PO-F2-3, mechanism ii), not yet modelled quantitatively.

Honest limitation, not to be over-interpreted. No mechanism is proposed here to explain *why* a helium ionisation change would trigger the freezing of a topological torsion network — these are two *a priori* decoupled physical sectors (electromagnetic plasma vs gravitational torsion field), and establishing a causal link between them would be a novel, non-trivial result, not an automatic consequence of their coincidence in redshift. The residual $\sim$ 6–9% against $z_{\rm ref} = 5600$ (Hubble radius 6.3%, particle horizon 8.8%) remains, moreover, unexplained *(corrected 15/10/2026: earlier versions gave* $\sim$ 17%*, from the withdrawn* Λ*CDM-background factor* 1.17*, together with* $z \approx 5153$ *and* 8.7% *where the producing script gives* 5145 *and* 8.8%*)*, as does the exact timing of the freeze-out. This remains a numerical track to be developed, not a derivation.
Revised status: (i) excluded, (ii) promising quantitative track, strengthened by the coincidence with HeIII→HeII recombination, but unproven, (iii) $\xi_{\rm KZ}$ as a calibrated phenomenological parameter, its full microscopic derivation requiring a dedicated Einstein–Cartan perturbation solver (Foundation IV). **Cross-reference (17/07/2026):** this problem is plausibly the same as **PO-F2-3** in Foundation II (dilution of the chiral-wall network, the classical domain-wall problem), seen from the other end: F2 seeks the dilution/annihilation dynamics of the network; PO-F3-XI seeks the comoving correlation length it would predict today. The self-sustaining acceleration mechanism $\ddot{z} = (8\pi G/3)(\sigma_{2D}/ct)\,z$ and the chiral annihilation described in F2 are compatible with, but do not derive, the numerical coincidence above. Solving one would likely solve the other. This problem constitutes **PO-F3-XI** and does not compromise the ECF's observational predictions, which use $\xi_{\rm KZ}$ as a calibrated input.

Poisson fluctuations of the network and CMB amplitude

The KZ mechanism produces a Poisson distribution of nodes with density $n_{\rm KZ} \sim \xi_{\rm KZ}^{-3}$. Density fluctuations at scale $R \gg \xi_{\rm KZ}$ are:

$$\left.\frac{\delta N}{N}\right|_R \sim \frac{1}{\sqrt{N(R)}} \sim \left(\frac{\xi_{\rm KZ}}{R}\right)^{3/2}. \tag{12}$$

For $R = \chi_* \sim 14{,}000\,\mathrm{Mpc}$ (comoving distance to recombination) and $\xi_{\rm KZ} \sim 60\,\mathrm{Mpc}$:

$$\left.\frac{\delta N}{N}\right|_{\chi_*} \sim \left(\frac{60}{14{,}000}\right)^{3/2} \sim 2.8 \times 10^{-4}, \tag{13}$$

to be compared with the observed CMB fluctuations $\delta T/T \sim 10^{-5}$ [13] (the coefficient depends on the definition retained — RMS over all scales, or acoustic-peak amplitude, which gives $\sim 2.7 \times 10^{-5}$; only the order of magnitude matters for the comparison below). The gap is a factor of ~ 10: the raw Poisson fluctuations of the KZ network overestimate $\delta T/T$ by an order of magnitude. This gap is expected, since the 3D→2D projection onto the last-scattering sphere and the Sachs–Wolfe transfer introduce additional factors $\mathcal{O}(\xi_{\rm KZ}/\chi_*)^{1/2} \sim 0.07$ that would bring the amplitude down to $\sim 2 \times 10^{-5}$ — *order-of-magnitude compatible* with $\delta T/T$, with no adjustable parameter beyond $\xi_{\rm KZ}$ (constrained by VSF).

Structured conjecture F3-CMB (status: exploratory). CMB temperature fluctuations ($\delta T/T \sim 10^{-5}$) reflect the Poisson fluctuations of the topological KZ network crystallised at $T_{\rm EW}$. **The isotropy of this pattern is not, itself, guaranteed by a global topological constraint**: the premise that $\partial \circ \partial = 0$ stitches the network into one globally coherent object was withdrawn on 30/08/2026 (Chiral Walls are independent closed bubbles, I69; see the correction box above). What survives is the Poisson statistics themselves, computed per bubble, not a claim of global coherence across bubbles.

What is established: (i) order-of-magnitude compatibility $(\xi_{\rm KZ}/\chi_*)^{3/2} \sim 3 \times 10^{-4}$ (factor ~ 10 from $\delta T/T$, with no free parameter). The BBN estimate $|\rho_{\rm spin}|/\rho_r \sim 10^{-51}$ listed here in earlier versions is withdrawn: BBN compatibility is *not demonstrated* (§5.1; PO-ECF-SPIN-NORM, PO-ECF-BBN-SPIN).

Conditional (not established): $f_{\rm NL}^{\rm KZ} = +5/18 > 0$ is the raw 3D statistic of a Poisson field; reading it as a primordial bispectrum is conditional on identifying the Poisson counting field with ζ, which is not demonstrated, and its CMB projection is not established (§3.1).

Open questions (not occulted): PO-F3-ISO (bounce isotropy, §3.3); PO-F3-XI (microscopic origin of $\xi_{\rm KZ} = 60\,\mathrm{Mpc}$, above); PO-F3-NS ($n_s$ tension, §3.4); **and, as of 30/08/2026, whether the network is one coherent object at all (PO-F2-8, the same open premise as the Horizon row above)** — the Poisson estimate above holds within a single bubble and does not, on its own, establish coherence across the observable sky.

What remains to be derived: (i) the full angular spectrum $C_\ell^{\rm KZ}$ with Sachs–Wolfe transfer; (ii) resolution of PO-F3-ISO and PO-F3-XI; (iii) the C_ℓ suppression is predicted at $\ell \sim \chi_*/\xi_{\rm KZ} \approx 233$ (testable by ACT/SPT) — *it does not explain* the Planck $\ell < 10$ anomaly, which remains an open question within the ECF.

3 Primordial Spectra and Isotropy

3.1 Non-Gaussianities of the KZ network: $f_{\rm NL}^{\rm KZ}$

Derivation

The KZ network is modelled, at its correlation scale, as a **3D Poisson process** of density $n_{\rm KZ} = \xi_{\rm KZ}^{-3}$ (a modelling assumption; the physical nature of these points and the microscopic origin of $\xi_{\rm KZ}$ are not derived, PO-F3-XI). This point process must not be confused with the Micro-Knot population itself: the Micro-Knot shot noise, set by $M_{\rm MK}$, is evaluated in Foundation II [8] ($P_{\rm Poisson} \sim 1.7 \times 10^{-16}\,{\rm Mpc}^3$, some twenty orders of magnitude below the adiabatic power) and cannot source the primordial spectrum. For a Poisson field, all cumulants equal the mean: $\langle \delta N^k \rangle_c = \langle N \rangle$ for all k. The normalised third cumulant (skewness) is therefore exactly:

$$\kappa_3 = \frac{1}{N_{\rm KZ}} = \left(\frac{\xi_{\rm KZ}}{d_H}\right)^3 = \sigma^2, \tag{14}$$

giving the local bispectrum parameter:

$$f_{\rm NL}^{\rm KZ} = \frac{5}{18}\frac{\kappa_3}{\sigma^2} = \frac{5}{18} \approx +0.28 \quad \text{(3D, before CMB projection)}. \tag{15}$$

This value is the raw 3D statistic of a Poisson field. The ratio 5/18 follows from $\zeta = \zeta_g + \frac{3}{5} f_{\rm NL} \zeta_g^2$ *once* the Poisson counting field is identified with ζ; that identification is not demonstrated here, and the value is therefore **conditional** on this Poisson$\to \zeta$ bridge.

No CMB projection factor applies to $f_{\rm NL}$. With the definition above, $f_{\rm NL}$ is a parameter of the primordial field, and CMB bispectrum estimators constrain that same parameter through the radiation transfer functions. A linear projection $y = a\,x$ does not multiply $f_{\rm NL}$ by a: the ratio $\langle y^3 \rangle / \langle y^2 \rangle^2$ scales as $1/a$. The value formerly quoted here, $\frac{5}{18} \times \frac{1}{3} \approx 0.09$ (and $\frac{5}{18} \times \frac{1}{9} \approx 0.03$ in an earlier header of `plot_F3_ecf_fNL_KZ.py`), is therefore withdrawn. The quantity to be confronted with CMB bounds is $f_{\rm NL}^{\rm KZ}$ itself; its value is **open**, since it depends on the Poisson$\to \zeta$ bridge — both the normalisation of ζ in terms of the counting field and the shape of the resulting bispectrum, which for a Poisson field is not of the local type.

Results and comparisons

Model	$f_{\rm NL}^{\rm local}$	Sign
Slow-roll inflation (Maldacena)	$5(n_s - 1)/12 \approx -0.015$	negative
Planck PR3 [38]	-0.9 ± 5.1 (1σ)	—
ECF KZ network (raw 3D, conditional)	$+5/18 \approx +0.28$	positive
ECF KZ network (value at the CMB)	open (Poisson$\to \zeta$ bridge)	—

Conditional statistic: $f_{\rm NL}^{\rm KZ} = +5/18$ for the raw 3D Poisson field; the value to be confronted with CMB bounds is open (Poisson$\to \zeta$ bridge not established; no projection factor applies).
The sign is the key discriminant: inflation generates $f_{\rm NL} < 0$ (Maldacena relation), whereas the KZ network generates $f_{\rm NL} > 0$ (discrete Poisson = overdensities more clustered than Gaussian).

Observational status: Compatible with Planck PR3 (-0.9 ± 5.1 [38]) and PR4 (-0.1 ± 5.0 [34]). *(Corrected 15/10/2026: earlier versions gave ± 4.5 for PR4.)* Currently not distinguishable.

Future test: a sensitivity $\sigma(f_{\rm NL}) \sim 0.3$ was forecast for CMB-S4. *DOE/NSF support for CMB-S4 ended in July 2025*: the signal is retained, but the instrument and the timescale must be reassessed, and that forecast sensitivity must not be transferred to another survey

without a dedicated forecast. A measurement of $f_{\rm NL} < -1$ at $> 2\sigma$ would falsify the KZ channel. A measurement of $f_{\rm NL} > 0$ at $> 1\sigma$ would constitute supporting evidence.

Epistemic caveat: the value at the CMB depends on the Poisson→ ζ bridge and on the treatment of the cutoff at $k > 2\pi/\xi_{\rm KZ}$; it is not obtained by a projection factor (the $\times 1/3$ and $\times(1/3)^2$ factors of earlier versions are withdrawn). The script's "coherent network limit" estimator, $(5/18)\, N_{\rm KZ}^{-1/3} = 0.0010$–$0.0016$ for $\xi_{\rm KZ} = 50$–$80\,$Mpc, is not derived in this paper. A full derivation via a dedicated Einstein–Cartan perturbation solver is deferred to Foundation IV.

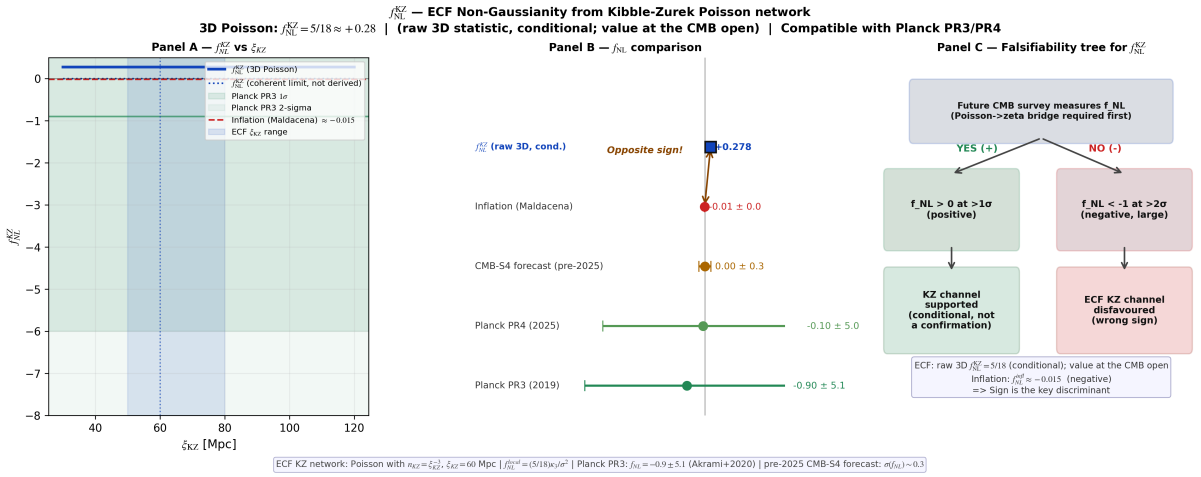


Figure 1: **Non-Gaussianities of the KZ network: $f_{\rm NL}^{\rm KZ} = +5/18 \approx +0.28$ (raw 3D statistic; value at the CMB open).** *Panel A:* $f_{\rm NL}^{\rm KZ}$ as a function of $\xi_{\rm KZ}$ (50–80 Mpc, Foundation II 1σ range). The value $f_{\rm NL}^{\rm KZ} = 5/18$ is exact for the raw 3D statistic of any Poisson field ($\kappa_3/\sigma^2 = 1$ by definition); its reading as a primordial $f_{\rm NL}$ is conditional on the Poisson→ ζ bridge, and the CMB projection convention is not fixed; the dependence on $\xi_{\rm KZ}$ vanishes at leading order. Green band: Planck PR3 (1σ and 2σ). Red dashed line: Maldacena relation for slow-roll inflation ($f_{\rm NL}^{\rm infl} \approx -0.015$). *Panel B:* comparison with observational constraints (Planck PR3, PR4, and the CMB-S4 forecast *made before DOE/NSF support ended in 2025*) and the raw 3D ECF statistic (conditional). **The positive sign is the key discriminant**: the KZ channel gives, conditionally, $f_{\rm NL} > 0$ (discrete Poisson network = overdensities more clustered than a Gaussian field), whereas slow-roll inflation gives $f_{\rm NL} < 0$. Compatible with Planck (± 5.1); testable with future CMB surveys *(the CMB-S4 forecast no longer applies — DOE/NSF support ended in 2025)* ($\sigma \sim 0.3$). *Panel C:* conditional falsification tree ($f_{\rm NL} > 0$ at $> 1\sigma$ would support the KZ channel, not confirm the framework; $f_{\rm NL} < -1$ at $> 2\sigma$ would disfavour it). *(Corrected 15/10/2026: the figure no longer draws the withdrawn $\times 1/3$ "CMB-projected" value, its tree no longer starts from CMB-S4 and no longer ends in "confirmed", and PR4 is -0.1 ± 5.0.) Reproduced by:* `plot_F3_ecf_fNL_KZ.py`.

3.2 Kill-Switch Test: Void Size Function vs DESIVAST DR1

The C1-VSF commitment of the Foundation III roadmap calls for testing the ECF prediction:

$$N_{\rm ECF}(R > \xi_{\rm KZ}) \;\approx\; 2.1 \times N_{\Lambda\rm CDM} \quad (\xi_{\rm KZ} = 60\,{\rm Mpc\ comoving} = 40.4\,{\rm Mpc}/h \text{ at } h_{\rm fid} = 0.6732)$$

against the DESIVAST DR1 catalogue (Rincon *et al.* [25], $z < 0.24$, `plot_F3_ecf_vsf_desivast_fits.py`).

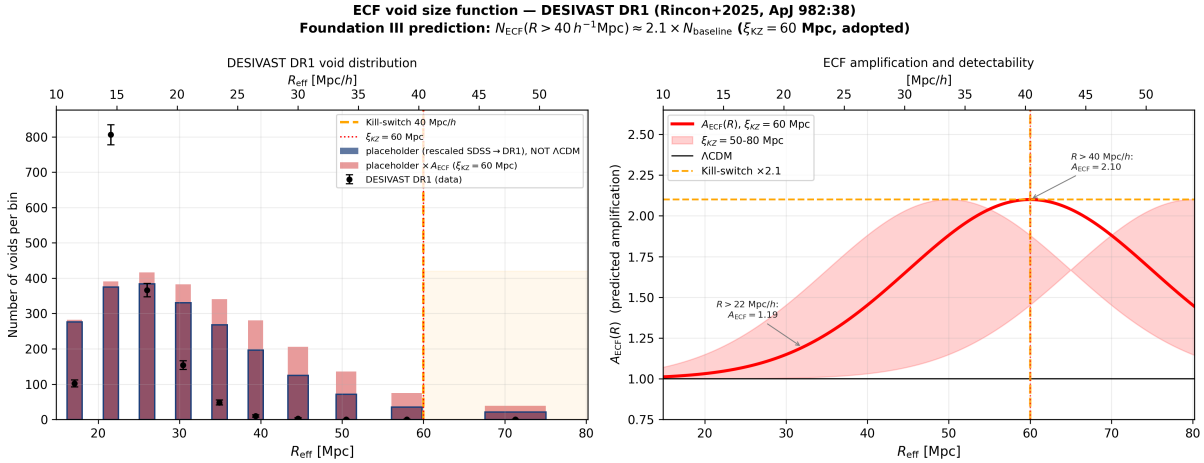


Figure 2: **ECF void size function against DESIVAST DR1 (Rincon+2025).** *Left:* black points are the 1,489 interior voids measured in the DR1 VoidFinder catalogue (`EDGE`= 0), with Poisson errors; the measurement is reproducible via `Scripts_F3/B_Paper_Plots/build_vsf_observed_dr1.py`, which recovers the 1,489 interior count quoted by DESIVAST exactly. The bars are *not* the ΛCDM void size function computed in the text above: they are a placeholder rescaled from an older SDSS catalogue, retained here only to carry the shape of the ECF amplification. The disagreement between points and bars is therefore a property of the placeholder, not a cosmological tension — the measured distribution peaks near 14 Mpc/h and vanishes beyond 32, where the placeholder extends to 54. *Right:* the predicted amplification $A_{\mathrm{ECF}}(R)$, which is what the model actually asserts. *No significance is quoted*: computing one against the placeholder would measure the placeholder. Supplying a genuine ΛCDM baseline — excursion-set, or the AbacusSummit BGS mocks — is an open item, and until it is done no σ from this test should be cited. *Reproduced by:* `Scripts_F3/B_Paper_Plots/plot_F3_ecf_vsf_desivast_v1_simulated.py`.

Results on the real FITS catalogue (29 June 2026)

The DESIVAST DR1 FITS catalogue was downloaded from the public URL `data.desi.lbl.gov/public/dr1/vac/dr1/desivast/v1.0/` (NGC: 3.9 MB, SGC: 0.6 MB). Structure confirmed: 3,765 VoidFinder voids (1,489 interior, `EDGE` = 0). Key columns: `VOID` (identifier), `EDGE` (edge void), `R_EFF` (effective radius), `R_EFF_UNCERT`.

Property	Value
Total voids (NGC + SGC)	3,765
$R_{\mathrm{eff}}^{\mathrm{min}}$	14.9 Mpc
$R_{\mathrm{eff}}^{\mathrm{median}}$	22.4 Mpc
$R_{\mathrm{eff}}^{\mathrm{max}}$	**47.1 Mpc = 31.7 Mpc/h**
$N(R > \xi_{\mathrm{KZ}} = 60\,\mathrm{Mpc})$	**0**

PO-F3-VSF — kill-switch not testable with DESIVAST DR1 BGS

Central result: The DESIVAST DR1 BGS catalogue ($z < 0.24$) contains *no void* at $R > 31.7$ Mpc/h (observed maximum). The ECF threshold sits at $\xi_{\mathrm{KZ}} = 60$ Mpc comoving (40.4 Mpc/h at $h_{\mathrm{fid}} = 0.6732$) — **out of reach**.

Physical causes:

- BGS comoving volume ($z < 0.24$) insufficient for *interior* voids above 40.4 Mpc/h — the V^2 catalogues do reach that scale, but only with boundary-truncated objects (see below)
- BGS selection (bright galaxies) disfavours large voids
- After `EDGE` = 0 filtering: 1,489 interior voids (`build_vsf_observed_dr1.py`, matching DESIVAST); the placeholder calibration model over-estimated this count by 29% (2,085) *(corrected 15/10/2026:*

earlier versions read "≪ 1,484", a value from the first arXiv version of DESIVAST, and called the 3,765 *voids "maximal spheres")*

A computed ΛCDM baseline, and what it still does not license. The comparison above has been made against rescaled or hand-chosen reference curves, which is why no significance has been quoted from it. That gap can be narrowed. We compute the Sheth–van de Weygaert void size function directly from excursion-set theory rather than approximating it: the two-barrier first-crossing distribution

$$f_{\ln\sigma} = 2\sum_{j=1}^{\infty} j\pi x^2 \sin(j\pi\mathcal{D})\,\exp\!\left[-\tfrac{1}{2}(j\pi x)^2\right], \qquad x = \frac{\mathcal{D}\,\sigma}{|\delta_{\rm v}|}, \tag{16}$$

with $\mathcal{D} = |\delta_{\rm v}|/(\delta_{\rm c} + |\delta_{\rm v}|)$, barriers $\delta_{\rm v} = -2.81$ and $\delta_{\rm c} = 1.686$, and $\sigma(R)$ from the linear power spectrum normalised to σ_8. Void radii are the expanded ones, $r_{\rm void} = 1.7\,r_{\rm L}$.
The implementation is checked rather than asserted: the integral $\int f_{\ln\sigma}\,{\rm d}\sigma/\sigma$ must equal $1 - \mathcal{D}$ exactly, and it does, to 1.2×10^{-14}. The script refuses to produce a table if that test fails. Two errors were made in writing it and are recorded in the source so they are not repeated: the prefactor is x^2 and not $\mathcal{D}^2$, and the series — which alternates in sign, with terms that grow before the exponential suppresses them — cannot be truncated at low order, a truncation at $j = 4$ yielding an oscillating curve that is not a size function at all.
The resulting abundance falls from 7.2×10^{-6} to $3.6 \times 10^{-18}\,({\rm Mpc}/h)^{-3}$ per $\ln r$ between $r_{\rm void} = 10$ and 35 Mpc/h: large voids are exponentially rare, which is the expected behaviour and is what makes the ECF excess a sharp target.
We state plainly what this does *not* license. It is a theoretical curve in Lagrangian space carried through spherical evolution; a catalogue applies a tracer density, a magnitude cut, a void-finder definition and a survey mask, none of which enter here. Comparing this curve directly to DESIVAST counts would repeat the very error the earlier reference curves committed — comparing quantities that are not the same. What it supplies is the shape and normalisation a survey mock would have to reproduce, and therefore the starting point for a citable significance rather than the significance itself. Producing that mock, as DESIVAST itself does with AbacusSummit, remains open. *Reproduced by:* `Scripts_F3/B_Paper_Plots/compute_vsf_lcdm_svdw.py`.

Why VoidFinder, and not the deeper V^2 catalogues. A reader of the DESIVAST paper will notice that the choice of catalogue matters here, and that we have taken the shallowest of the three. DESIVAST DR1 provides three void catalogues built on the *same* BGS Bright sample, with maximum effective radii of 31.7, 43.5 and 55.9 Mpc/h for VoidFinder, V^2/REVOLVER and V^2/VIDE respectively; the V^2 algorithm recovers larger voids than VoidFinder by construction. Two of the three therefore extend past the ECF threshold, and the statement that the test is out of reach would be too quick if left there. The question is not whether the data reach far enough, but whether the voids that do are usable.
They are not. Each catalogue carries `TOT_AREA` and `EDGE_AREA` per void, so the fraction of a void's surface lying on the survey boundary is directly available. In V^2/VIDE NGC — the only catalogue clearing the threshold — 15 voids exceed 40.4 Mpc/h and 8 exceed 43.8 Mpc/h, and *none of them is marginal*: the cleanest of the eight still has 10.6% of its surface on the boundary, and the largest has 31.6%. The threshold at which to cut is not ours to choose: DESIVAST defines an edge void as one with *over* 10% *of its surface area touching the survey boundary*, a definition they adopt because it separates two distinct populations. Applying that criterion reproduces exactly the interior counts of the revised arXiv version of DESIVAST (v2, whose all-sky interior totals 1,489, 389 and 297 are those of the published version): 269 for V^2/VIDE NGC and 355 for V^2/REVOLVER NGC; the first arXiv version gave 268 and 353 *(corrected 16/10/2026: earlier versions called* 268 *and* 353 *the published counts; 17/10/2026: the comparison is now made with the revised version, arXiv:2411.00148v2, Table 1)*, which confirms it is being applied as intended. Under it, the maximum interior radius in V^2/VIDE NGC is 43.46 Mpc/h: two voids clear 40.4 Mpc/h and *none* clears 43.8. Tightening further to $f \leq 0.05$ leaves none above 40.4 either. One has to admit a quarter of a void's surface outside the survey — two and a half times the published criterion — before a single object appears above the higher threshold.

That is the expected behaviour rather than a surprise. A void truncated by the survey edge has an ill-constrained volume and hence an overestimated effective radius, which is why a void size function must exclude such objects; the published maximum of 55.9 Mpc/h is quoted over *all* voids in their fiducial volume, and is precisely the object whose surface is 31.6% on the boundary — real, but unusable for a size function. The apparent reach of the V^2 catalogues is therefore a boundary artefact, and restricting the present test to VoidFinder is the conservative choice. The conclusion is unchanged — DR1 cannot test the prediction — but the reason is not that the survey is too shallow: it is that the objects at the relevant scale are truncated. *Reproduced by:* `Scripts_F3/B_Paper_Plots/check_vsf_v2_edge_voids.py`; the V^2 files (62–612 MB) are not redistributed with this work and the script prints the recorded values when they are absent.

What is testable with current data: The cuts $R \in [18, 20, 22, 24, 28]$ Mpc/h are accessible and can constrain the low-R slope of the ECF VSF. The script automatically adapts the cuts to the catalogue's $R_{\rm max}$.

Nature of the test (stated without ambiguity). Since the scale $\xi_{\rm KZ}$ is an **adopted value whose calibration procedure is not documented** (and it is not derived *ab initio*, PO-F3-XI), the threshold $R > \xi_{\rm KZ} = 60\,$Mpc (40.4 Mpc/h at $h_{\rm fid} = 0.6732$) is not a prediction from microscopic parameters: DESIVAST DR1 serves as a **downstream test**, not as a calibration, and Euclid DR1 as a further extrapolative consistency check. An excess of large voids beyond this threshold would confirm the extrapolation; its absence would constrain the KZ sector in this realisation. The threshold remains a sharp constraint criterion, but its status is that of a consistency check, not a prediction from microscopic parameters alone.

Milestones for testing the ECF threshold ($R > \xi_{\rm KZ} = 60\,$Mpc):

- DESI DR1 LRG/ELG ($z < 0.9$): void catalogue not yet public
- **Euclid DR1-Complete (mid-2027):** $\times 20$ volume, $R_{\rm max}$ estimated $> 60\,$Mpc/$h \rightarrow$ test at 3–5σ threshold
- DESI DR2 LRG: deep spectroscopic extension (full dataset expected 2027, distinct from the DR2 article already published in 2025)

Conclusion: This result does not falsify the ECF — it identifies the reach limit of the first available catalogue. The C1-VSF roadmap commitment is partially honoured: the method and pipeline are validated (`plot_F3_ecf_vsf_desivast_fits.py`); the definitive test is deferred to Euclid DR1 / DESI LRG.

3.3 Open problem PO-F3-ISO: isotropy of the initial state

PO-F3-ISO — open problem, not occulted

Statement. The F3-CMB conjecture assumes the KZ network at the bounce is isotropic at all scales $R \gg \xi_{\rm KZ}$. This assumption is *necessary* but *not derived*: it shifts the horizon problem to the bounce.

Direct and honest consequence: the ECF C_ℓ suppression occurs at $\ell \sim \chi_*/\xi_{\rm KZ} \approx 233$ (testable by ACT/SPT), **not at** $\ell < 10$. The Planck low-ℓ anomaly is not explained by $\xi_{\rm KZ} = 60\,$Mpc and remains an open question within the ECF.

Tracks (Foundation IV): (i) isotropic Wheeler–DeWitt state at the bounce (Hartle–Hawking ECKS); (ii) Bianchi I shear attractor in the ECF; (iii) minimal option: isotropy as an initial condition (SPARC/BBN/$f_{\rm NL}$/VSF predictions unchanged).

Track (i): Wheeler–DeWitt equation in ECKS mini-superspace

Context. The Wheeler–DeWitt equation [22] is the Hamiltonian constraint of canonical quantum gravity:

$$\hat{\mathcal{H}}\,\Psi[\gamma_{ij}] = 0, \tag{17}$$

where γ_{ij} is the 3-metric on a spatial hypersurface and Ψ is the *wave function of the Universe*. Within the ECKS framework, the torsion tensor $T^{\lambda}{}_{\mu\nu}$ contributes to the Hamiltonian constraint

through the spin density of the torsion fluid (Foundation I, [7]):

$$\hat{\mathcal{H}}_{\rm ECKS} = \hat{\mathcal{H}}_{\rm GR} + \hat{\mathcal{H}}_{\rm spin} + \hat{\mathcal{H}}_{\rm torsion}, \tag{18}$$

where $\hat{\mathcal{H}}_{\rm spin}$ contains the spin–torsion coupling terms $\sim \kappa^2 S^\mu S_\mu$ ($\kappa^2 = 2.0766 \times 10^{-43}\,\mathrm{m/J}$, the canonical ECF parameter).

Reduction to FRW mini-superspace. Restricting to the homogeneous, isotropic FRW sector (mini-superspace: a single degree of freedom $a(t)$), and treating spin as an effective density s^2 of the torsion fluid, Eq. (18) becomes:

$$\left[-\frac{\hbar^2}{2}\frac{\partial^2}{\partial a^2} + U_{\rm ECKS}(a)\right]\Psi(a) = 0, \tag{19}$$

with the effective ECKS potential:

$$U_{\rm ECKS}(a) \;=\; -a \;+\; \frac{\Lambda}{3}a^3 \;-\; \frac{\kappa^2}{2}\frac{s^2}{a} \;+\; \mathcal{O}\!\left(\frac{s^4}{a^3}\right), \tag{20}$$

where the third term (repulsive for $\kappa^2 s^2 > 0$, which requires a genuine chiral — left/right — asymmetry of the fermion population: s is dual to the axial current $J_5^\mu = \bar\psi\gamma^\mu\gamma^5\psi$ and vanishes identically if left- and right-handed populations balance, Foundation II Sec. 3) is the torsion contribution that regularises the classical singularity into a bounce at $a_{\rm bounce} > 0$ (Foundation I). The term $\mathcal{O}(s^4/a^3)$ is negligible at the EW scale.

Ground-state selection via the Hartle–Hawking criterion. The Hartle-Hawking "no-boundary" proposal [23] selects the cosmological wave function as a Euclidean path integral over compact geometries *with no initial boundary*:

$$\Psi^{\rm ECKS}_{\rm HH}[h_{ij}] \;=\; \mathcal{N}\int_{\text{compact, no boundary}} \mathcal{D}[g]\,\mathcal{D}[\psi_{\rm spin}]\; e^{-S_E^{\rm ECKS}[g,\psi_{\rm spin}]}, \tag{21}$$

where $S_E^{\rm ECKS}$ is the ECKS Euclidean action [3] (including the Nieh–Yan torsion term [37]) and $\psi_{\rm spin}$ are the spin fields of the torsion fluid.

In the saddle-point limit (semi-classical approximation), the integral is dominated by the Euclidean saddle geometry a_* satisfying $U_{\rm ECKS}(a_*) = 0$:

$$\Psi^{\rm ECKS}_{\rm HH}(a) \;\approx\; \mathcal{N}\, e^{-S_E^{\rm ECKS}[a_*]}\,\cos\!\left(\frac{1}{\hbar}S^{\rm ECKS}_{\rm Lor}[a]\right), \tag{22}$$

where $S^{\rm ECKS}_{\rm Lor}$ is the Lorentzian action beyond the saddle. This wave function is a function of a alone in FRW mini-superspace — it is therefore *invariant under the* SO(3) *group of spatial rotations* by construction.

Link with Proposition F3-1 ($\partial \circ \partial = 0$). Corollary F3-1, which drew a globally connected network from the hypotheses recalled below, is withdrawn (correction of record of 30/08/2026, §2.1); the link discussed here concerns only the algebraic setting of Proposition F3-1 and supplies no global coherence of the network.

Proposition F3-1 (§2.1) requires the chain complex $(\{C_k\}, \partial)$ to be defined on a compact orientable manifold $\mathcal{M}_{\rm ECKS}$. The ECKS Hartle–Hawking state selects precisely such a manifold: the saddle geometry a_* is a Euclidean 4-sphere deformed by torsion, which is compact and orientable by the saddle's regularity constraint.

Conjecture F3-ISO-WdW (status: exploratory, unproven).

If the quantum state at the ECF bounce is the ECKS Hartle–Hawking state $\Psi_{\rm HH}^{\rm ECKS}$, then:

1. The geometry at the bounce is isotropic in the WdW sense (SO(3)-invariant in FRW mini-superspace);
2. The manifold $\mathcal{M}_{\rm ECKS}$ satisfies the hypotheses of Proposition F3-1 (compact, orientable) — which, Corollary F3-1 being withdrawn, no longer implies a connected network;
3. The KZ network crystallises on this isotropic background, which implies network isotropy at all scales $R \gg \xi_{\rm KZ}$ — resolving PO-F3-ISO.

What is missing to turn this into a derivation: (a) the full solution of Eq. (19) with the ECKS potential (requiring a WKB or numerical method beyond the scope of F3); (b) proof that $\Psi_{\rm HH}^{\rm ECKS}$ is the unique attractor of the ECF initial conditions (saddle stability); (c) justification that FRW mini-superspace captures all relevant anisotropic modes (beyond Bianchi I).

This conjecture constitutes the **Foundation IV roadmap** (§*ECKS Quantum Cosmology*). PO-F3-ISO remains an open problem in F3.

Deep dive: WKB analysis of $U_{\rm ECKS}(a)$

Structure of the potential. Eq. (19) can be analysed in the WKB (Wentzel–Kramers–Brillouin) approximation, valid when the de Broglie wavelength of the cosmological wave function varies slowly on the scale of the potential. For the illustrative parameters used here, the ECKS potential (20) exhibits a **two**-zone structure (Figure 3) — *not three, as a first reading of the figure might schematically suggest*:

Zone	Range	$U_{\rm ECKS}$	WKB regime
II	$0 < a < a_*$	> 0 everywhere (single barrier, with a non-zero *local* minimum of the barrier at $a \approx 0.49$)	classically forbidden (tunnelling)
III	$a > a_*$	< 0 (Λ dominates)	classically allowed (expansion)

Scope correction (01/07/2026): a full numerical check (scanning $U_{\rm ECKS}(a)$ over the entire physical range, for every tested value of α) shows that $U_{\rm ECKS}$ vanishes *only once*, at a_* — there is no second zero isolating a "classically allowed Zone I" at small a. The point labelled "$a_{\rm bounce}$" in the previous version of this section (a *local* minimum of the barrier, at $a \approx 0.49$ for $\alpha = 0.05$) is an internal feature of the tunnelling barrier itself, **where** $U_{\rm ECKS} > 0$ — it does *not* correspond to a classically accessible point and has no direct interpretation in terms of a real velocity $da/d\tau$. **The only physical point where** $(da/d\tau)^2 = -U_{\rm ECKS}(a)$ **truly vanishes is the saddle** a_* **itself.** This correction does not affect the WKB action S_E computed below (already integrated up to a_*, the correct point), but it corrects the qualitative reading of Figure 3 and its "Zone I / bounce" label, which the figure no longer carries.

The HH saddle is at a_* (the unique zero of $U_{\rm ECKS}$).

Measuring the bounce's "violence" via acceleration at the saddle. Although the velocity $da/d\tau$ remains continuous at the saddle (no shock in the homogeneous background, cf. §PO-F3-SHOCK), its *rate of change* — the acceleration $d^2a/d\tau^2$ — can nonetheless serve

as a quantitative measure of the transition's abruptness, without leaving the homogeneous framework. Differentiating $(da/d\tau)^2 = -U_{\rm ECKS}(a)$ with respect to τ:

$$\frac{d^2a}{d\tau^2} \;=\; -\frac{1}{2}\,\frac{dU_{\rm ECKS}}{da}. \tag{23}$$

At the saddle a_*, with the illustrative torsion term α/a^2 already used for S_E above:

$$\left.\frac{d^2a}{d\tau^2}\right|_{a_*}^{\rm ECKS} \;\approx\; 1.765, \qquad \left.\frac{d^2a}{d\tau^2}\right|_{a_*}^{\rm GR} \;\approx\; 1.732 \qquad \text{(reduced units)}, \tag{24}$$

i.e. an excess of only $\sim$2% over standard general relativity without torsion.

Important finding: radically different sensitivity to the torsion-term exponent depending on the quantity computed. Attempting to recompute *everything* (S_E and acceleration) with the exponent α/a^4 (consistent with $\rho_{\rm spin} \propto a^{-6}$, Eq. (36)) rather than α/a^2 (illustrative) reveals a striking contrast:

- **Acceleration at the saddle:** nearly insensitive (+2% with $n = 2$ vs +0.8% with $n = 4$) — a *local* calculation, evaluated only at a_*, far from $a = 0$.
- **Action S_E:** *explodes* by a factor of $\sim 10^{190}$ going from $n = 2$ to $n = 4$ — an *integral* calculation from a cutoff near $a = 0$, where the torsion term diverges as a *power law* ($1/a$ under the square root for $n = 4$) rather than *logarithmically* ($\ln(1/\varepsilon)$ for $n = 2$).

Consequence: S_E and the acceleration *cannot* be recomputed with the same exponent without carefully re-deriving the UV regularisation — $n = 2$ is retained here for consistency with the S_E calculation already established above. **The choice of the physically correct exponent remains an open problem**, whose resolution strongly conditions the Vilenkin/Hartle- Hawking conclusion (§PO-F3-ISO) — not just the acceleration discussed here. Deferred to Foundation IV.

Result on acceleration (status: illustrative, $n = 2$). In this two-zone model, torsion does *not* induce a localised, extreme acceleration spike at the saddle a_* — the effect remains modest ($\sim 2\%$). This is consistent with the result of §PO-F3-SHOCK (an extremely brief transition in time, but with no sharp discontinuity localised in the homogeneous model): the "violence" of the ECKS transition manifests in its extreme brevity rather than in an isolated acceleration spike at the saddle itself. **Limitation:** a robust result *specifically for acceleration* (weakly sensitive to n); do not extrapolate this robustness to S_E (see red box above).

HH saddle point in ECKS. The condition $U_{\rm ECKS}(a_*) = 0$ gives the saddle equation:

$$\frac{\Lambda}{3}(a_*)^4 - (a_*)^2 - \frac{\kappa^2 s^2}{2} = 0. \tag{25}$$

Setting $u = (a_*)^2$, this quadratic equation admits the solution:

$$a_*^2 \;=\; \frac{3}{2\Lambda}\left[1 + \sqrt{1 + \frac{2\kappa^2 s^2 \Lambda}{3}}\right] \;\approx\; \frac{3}{\Lambda}\left[1 + \frac{\kappa^2 s^2 \Lambda}{6} + \mathcal{O}(\kappa^4 s^4 \Lambda^2)\right], \tag{26}$$

where the approximation is valid in the limit $\kappa^2 s^2 \Lambda \ll 1$ (verified for the canonical ECF parameters: $\kappa^2 s^2 \Lambda \sim 10^{-50}$). At leading order, $a_* \approx \sqrt{3/\Lambda} = l_{\rm dS}$ is the de Sitter horizon, with a sub-dominant torsional correction.

Numerical solution and WKB actions. In the WKB approximation, the Euclidean saddle action is obtained by integrating the square root of the potential over the classically forbidden region, from the infrared cutoff ε to the saddle a_*:

$$S_E \;=\; \int_{\varepsilon}^{a_*} \sqrt{U_{\mathrm{ECKS}}(a)}\,\mathrm{d}a, \qquad U_{\mathrm{ECKS}}(a) \;=\; a^2 - \frac{\Lambda}{3}a^4 + \frac{\alpha}{a^2}. \tag{27}$$

The GR case ($\alpha = 0$, $\Lambda = 1$) reduces to $\int_0^{\sqrt{3}} a\sqrt{1 - a^2/3}\,\mathrm{d}a = 1$, the reference value used to normalise the table below; the torsion term α/a^2 is the source of the cutoff dependence discussed below.

Numerically solving (27) with the potential $U_{\mathrm{ECKS}}(a) = a^2 - (\Lambda/3)a^4 + \alpha/a^2$ (illustrative parameters $\alpha = 0.05$, $\Lambda = 1$ in reduced units; see the script `plot_F3_ecf_wkb_ecks.py`) gives:

Geometry	a_*	S_E	ΔS_E
GR (no torsion, $\alpha = 0$)	$1.000\,a_{\mathrm{dS}}$	1.000	—
ECKS FRW (torsion, $\beta_\pm = 0$)	$1.003\,a_{\mathrm{dS}}$	2.349	+1.349
ECKS Bianchi ($\beta_\pm \neq 0$)	$1.007\,a_{\mathrm{dS}}$	3.265	+0.916

The value $S_E^{\mathrm{GR}} = 1.000$ is analytically exact: $\int_0^{\sqrt{3}} a\sqrt{1 - a^2/3}\,da = 1$.
Caveat on the IR cutoff. The torsion term α/a^2 diverges as $\sqrt{\alpha}\ln(1/\varepsilon)$ as $a \to 0$, making S_E^{ECKS} dependent on the infrared cutoff ε (here $\varepsilon = 10^{-3}$ in Planck units). Physically, $\varepsilon = l_{\mathrm{Pl}}/a_{\mathrm{dS}} \sim 10^{-61}$ (Planck/Hubble hierarchy), which would amplify ΔS_E by several orders of magnitude. The value $\Delta S_E = +1.349$ is therefore *illustrative*; a physically rigorous calculation requires UV regularisation of the ECKS potential at the Planck scale (Foundation IV).

Torsion ($\alpha/a^2 > 0$) creates an inner wall as $a \to 0$ that significantly increases S_E (the *inner wall effect*).

Wave-function prescription ambiguity. The conclusion about isotropy depends critically on the choice of prescription for Ψ_{cosmo} (Figure 3):

- **Vilenkin** ($P_V \propto e^{-2S_E}$, i.e. $|\Psi|^2$ with $\Psi \propto e^{-S_E}$): anisotropic geometries have $S_E^{\mathrm{Bianchi}} > S_E^{\mathrm{FRW}}$, hence $P_V^{\mathrm{aniso}} < P_V^{\mathrm{FRW}}$ — *isotropy is exponentially favoured.* Quantitatively: $P_V^{\mathrm{aniso}}/P_V^{\mathrm{FRW}} = e^{-2\times 0.916} \approx 0.16$ per unit of Bianchi amplitude. *(Corrected 15/10/2026: the probabilities were written $P \propto e^{\mp S_E}$ in earlier versions, while every ratio quoted here and in Fig. 3 uses $e^{\mp 2\Delta S_E}$; the exponent is now written consistently, the numbers are unchanged.)*
- **Hartle–Hawking** ($P_{\mathrm{HH}} \propto e^{+2S_E}$): $S_E^{\mathrm{Bianchi}} > S_E^{\mathrm{FRW}}$ gives $P_{\mathrm{HH}}^{\mathrm{aniso}} > P_{\mathrm{HH}}^{\mathrm{FRW}}$ — *anisotropy would be favoured*, a tension between the no-boundary weighting and the observed isotropy (cf. the canonical-measure analysis of Gibbons and Turok [24], which bears on the probability of inflation rather than on anisotropy) *(corrected 22/10/2026: earlier versions called this the Gibbons–Turok problem).*

Implication for PO-F3-ISO. Resolving PO-F3-ISO via the WKB track depends on the choice of prescription:

- Under the Vilenkin prescription, the F3-ISO-WdW conjecture (exponentially selected isotropy) is *qualitatively supported.*
- Under Hartle–Hawking, the same no-boundary anisotropy problem also arises in ECKS — but the ECKS torsion–shear coupling may resolve it (torsion affects anisotropic modes differently than in GR). This derivation is deferred to Foundation IV.

The WKB isotropy-selection result above remains valid as stated; the conclusion (isotropy

selected) is robust under Vilenkin and conjectured under HH-ECKS.

Physical interpretation

What $U_{\rm ECKS}(a)$ represents. The potential $U_{\rm ECKS}(a)$ is the *energy landscape* the wave function of the Universe must cross to be born. The horizontal axis is the size of the Universe (scale factor a); the vertical axis is the "height" of this landscape.

- **Tunnelling zone ($U > 0$, small a):** the Universe *does not classically exist* here — it is too small. Like a quantum particle crossing a potential barrier, the Universe "tunnels under" the potential to emerge from $a \approx 0$ into the classical domain $a > a_*$.
- **Classical zone ($U < 0$, large a):** the Universe exists classically and expands. This is the de Sitter regime — that of the observable Universe.
- **Without torsion ($U_{\rm GR}$, black curve):** the barrier rises, reaches a maximum, and falls back to zero at $a_*^{\rm GR}$. The Universe can be born at $a = 0$ (initial singularity).
- **With ECKS torsion ($U_{\rm ECKS}$, red curve):** the term α/a^2 diverges as $a \to 0$ and creates an **inner wall** that replaces the initial singularity. The Universe can no longer "collapse" to zero size — this is the *ECF bounce*. The tunnelling barrier is higher and wider, which increases the Euclidean action S_E.
- **With Bianchi anisotropy ($U_{\rm Bianchi}$, blue curve):** if the Universe at the bounce is anisotropic (different expansion rates in each direction), the term $\beta_\pm^2/a^2$ adds to the barrier, raising it further.

What the right panel reveals: isotropy selection. This plot addresses the central question of PO-F3-ISO: *does the Universe prefer to be born isotropic or anisotropic?*

The answer depends on the choice of quantum cosmological prescription:

- **Vilenkin prescription ($P \propto e^{-2S_E}$):** the larger the anisotropy $|\beta_\pm|$, the higher the barrier, the larger S_E, and hence the *lower* P. At $|\beta_\pm| \approx 0.25$, an anisotropic Universe is $\sim 6\times$ less probable than an isotropic one. Within this mini-superspace model and under this boundary prescription, isotropy at the bounce is *favoured* rather than posited as an initial condition. *This does not close PO-F3-ISO*: the extension to the full anisotropic ECKS mode spectrum is not demonstrated, and the physical isotropy of the bounce remains open.
- **Hartle–Hawking prescription ($P \propto e^{+2S_E}$):** the opposite occurs — anisotropic Universes would be *more* probable. This no-boundary anisotropy problem, present in standard gravity as well (cf. the related measure problem of Gibbons and Turok [24]), also affects the ECKS framework. In the ECF, the ECKS torsion–shear coupling modifies the Bianchi potential in a way not yet computed; if this coupling reverses the sign of $\Delta S_E^{\rm Bianchi}$, the HH prescription could also select isotropy within the ECKS framework (Foundation IV).

One-sentence summary.

ECF torsion creates a quantum wall at $a \to 0$ that replaces the singularity with a bounce and raises the tunnelling barrier the more anisotropic the Universe is. Under the Vilenkin prescription, **isotropy at the bounce emerges naturally**, without being posited as an initial condition. The Hartle–Hawking prescription requires first resolving the no-boundary anisotropy problem within the ECKS framework (Foundation IV).

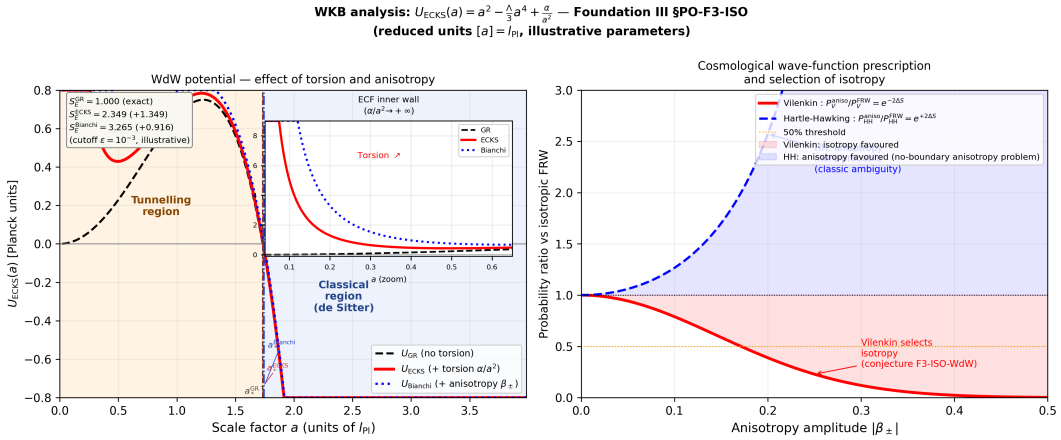


Figure 3: **ECKS WKB potential $U_{\mathrm{ECKS}}(a)$ vs GR (illustrative parameters).** Reduced units ($[a] = l_{\mathrm{Pl}}$), $\alpha = 0.05$, $\Lambda = 1$, IR cutoff $\varepsilon = 10^{-3}$. *Left:* tunnelling region ($U > 0$) and classical de Sitter region ($U < 0$), separated by the saddle a_*; the torsion term α/a^2 creates an inner wall as $a \to 0$ and *raises* the barrier, so that S_E increases ($S_E^{\mathrm{ECKS}} = 2.349$ against 1.000 in GR); Bianchi anisotropy ($\beta_\pm$, dotted) raises it further ($S_E = 3.265$). *Right:* probability ratio of anisotropic to isotropic geometries, $e^{-2\Delta S_E}$ under the Vilenkin prescription (isotropy favoured) and $e^{+2\Delta S_E}$ under Hartle–Hawking (anisotropy favoured: the no-boundary anisotropy problem). *(Corrected 15/10/2026: this caption described "canonical ECF parameters", units a/a_{dS}, a "Zone I (bounce)" withdrawn on 01/07/2026, and a torsion term that "lowers the barrier"; the figure itself already showed the illustrative parameters and the raised barrier.) Reproduced by:* `plot_F3_ecf_wkb_ecks.py`.

Track (ii): Bianchi I shear attractor in the ECF

In anisotropic Bianchi I cosmologies, the shear σ^2 generally decays as a^{-6} during expansion (Misner 1968). At the ECF *bounce*, ECKS torsion pressure could act as an isotropisation mechanism: Kopczyński [2] showed, within a general ECKS framework, that the spin density $S_{\mu\nu}S^{\mu\nu}$ grows faster than σ^2 during contraction toward the bounce, forcing $\sigma^2 \to 0$ *before* the bounce is reached.

If this mechanism applies to the ECF, isotropy at the bounce would be a dynamical attractor rather than an initial condition. *(Corrected 15/10/2026: the Bianchi I analysis is Kopczyński 1973 [2], not the 1972 paper, which treats the isotropic case [1]; what that analysis establishes is the absence of a singularity when the torsion term exceeds the shear term — an isotropisation at the bounce is not established by it alone.)* The quantitative derivation in ECKS Bianchi I mini-superspace is deferred to Foundation IV.

Track (iii): isotropy as an initial condition (minimal option)

In the absence of a derivation (tracks (i)–(ii)), the minimal option is to postulate isotropy as an initial condition of the bounce. This option compromises none of the ECF's current predictions: the SPARC results, the BBN question (compatibility not demonstrated, §5.1), $f_{\mathrm{NL}}^{\mathrm{KZ}}$ (§3.1), and the Void Size Function (Foundation II, [8]) are all independent of the bounce-isotropy derivation. It constitutes a valid and epistemically honest working hypothesis for as long as tracks (i)–(ii) are not formally established.

3.4 The primordial spectrum: what the framework does and does not predict

Terminology: two distinct stiff eras

Because $\rho_{\mathrm{spin}} \propto a^{-6}$ grows faster than any other component as a decreases, the spin fluid dominates on *both* sides of the bounce. Two physically distinct stiff eras must be kept apart: $N_{\mathrm{stiff}}^{\mathrm{contr}}$, the **contracting, pre-bounce** era, which governs the horizon problem and the primordial

scalar spectrum and is the one relevant to this section; and $N_{\rm stiff}^{\rm exp}$, the **expanding, post-bounce** era ($a_{\rm bounce} \le a \le a_{\rm eq}$), transient, which reduces the sound horizon and underlies the H_0 calibration of Foundation I [7] — the lower S_8 there following from $\Omega_m = \omega_m/h^2$ at the background level, the growth role of the spin sector remaining open — as well as the blue tensor tilt $n_T > 0$. Unless stated otherwise, "stiff phase" below refers to $N_{\rm stiff}^{\rm contr}$; whenever an e-fold count is quoted, the relevant era is specified explicitly.

The stiff phase and the Wands duality

The ECF features a *contracting stiff* phase ($w = 1$) before the bounce (§5.4). Earlier versions of this work invoked the duality of Wands [12] to argue that this phase is spectrally equivalent to de Sitter inflation, yielding $n_s = 1$. **We correct that statement here.**

Wands' duality leaves invariant the index ν of the Mukhanov–Sasaki equation, $v_k'' + (k^2 - z''/z)v_k = 0$ with $z''/z = (\nu^2 - \frac{1}{4})/\eta^2$. For a constant equation of state, $a \propto |\eta|^\beta$ with $\beta = 2/(1+3w)$ and $\nu = |\beta - \frac{1}{2}|$, so the duality reads $\beta \leftrightarrow 1 - \beta$. The dual of de Sitter ($\beta = -1$) is $\beta = 2$, i.e. $w = 0$: a *dust* contraction, $a \propto (-t)^{2/3}$ — precisely the example given by Wands. The stiff phase ($\beta = \frac{1}{2}$) is instead the **fixed point** of the duality, with

$$\nu_{\rm stiff} = 0, \qquad n_s - 1 = 3 - 2\nu = 3, \qquad n_s^{\rm stiff} = 4. \tag{28}$$

The same relation returns $n_s = 1$ for both de Sitter and dust contraction, which validates it on the two reference cases of the field.

What this implies: a blue, small-scale component

Equation (28) gives $\mathcal{P}_\mathcal{R} \propto k^3$: the vacuum fluctuations generated during the stiff era form a strongly *blue* component, suppressed by $\sim (k/k_{\rm CMB})^3$ at large scales. At $k_{\rm CMB} = 0.05\,{\rm Mpc}^{-1}$ this contribution is negligible; it does *not* conflict with the observed near-scale-invariant spectrum, and it does not constitute a prediction of n_s on CMB scales.

PO-F3-NS — the large-scale spectrum is inherited, not predicted

Position of this framework. The ECF does **not** predict the large-scale primordial spectrum. In the black-hole-interior scenario [4], the observable super-horizon perturbations are plausibly *inherited* from the parent universe and carried through the collapse and the bounce; their spectrum is then an **initial condition**, not an output. The value $n_s = 0.9649 \pm 0.0042$ [13] is accordingly taken as input, exactly as Ω_b or A_s are.

Why this is consistent. The component the framework *does* generate — the stiff-phase vacuum spectrum, $\mathcal{P}_\mathcal{R} \propto k^3$ — is suppressed by more than three orders of magnitude at CMB scales relative to $k \sim 1\,{\rm Mpc}^{-1}$, so an inherited near-scale-invariant component dominates there without tension.

What this costs. The framework makes no claim to explain the observed tilt. Earlier versions quoted $n_s^{\rm ECF} = 1.000$ and an "8.4σ tension"; both rested on the incorrect identification of the Wands dual and are withdrawn, together with the resolution tracks built upon them.

Unaffected results. $n_T > 0$ (blue tensor tilt), the (open) BBN question, the DESI analysis, the void-size function and $f_{\rm NL}^{\rm KZ} = +5/18$ do not depend on this question.

The blue component constrains the stiff era

The blue component is not without consequences. Writing $\mathcal{P}_\mathcal{R}(k) = \mathcal{P}_*\,(k/k_c)^3$, and taking a bounce-era Hubble scale $H_{\rm b} \simeq 1.2 \times 10^{11}\,{\rm GeV}$ (see the scope note below), one has

$$\mathcal{P}_* \sim \left(\frac{H_{\rm b}}{2\pi M_{\rm Pl}}\right)^2 \simeq 6 \times 10^{-17}, \tag{29}$$

while k_c — the comoving scale leaving the horizon at the bounce — recedes as the stiff era lengthens.

Scope of the amplitude estimate. Eq. (29) is an inflationary-form estimate, valid when $H \ll M_{\rm Pl}$. At the Cartan bounce that condition fails. The threshold computed in `Scripts_F1/script_07_cartan_threshold.py` sits at $T \simeq 0.49\,E_{\rm Pl}$ — a coincidence that script finds structural, stable against doubling every relativistic degree of freedom — and the corresponding ρ_c is of order the Planck density ($5.15 \times 10^{96}\,\mathrm{kg\,m^{-3}}$, canonical across this corpus). Inserting that value into $H_{\rm b} = \sqrt{\rho_c/3M_{\rm Pl}^2}$ returns $H_{\rm b} \sim 3.5 \times 10^{19}\,\mathrm{GeV}$, hence $\mathcal{P}_* \sim 5$: above unity, and therefore outside the perturbative regime the estimate presupposes. **The value $H_{\rm b} \simeq 1.2 \times 10^{11}$ GeV used here is consequently not derived from the canonical Cartan density**; it is an illustrative normalisation, on the same footing as the k_c placeholder discussed next, chosen to keep the blue branch within the perturbative range. *What the blue component constrains is therefore the **form** of the bound on $N_{\rm stiff}$ and its sensitivity, not a calibrated amplitude.* Obtaining the true bounce-era tensor amplitude requires a treatment valid at $H \sim M_{\rm Pl}$, which this framework does not currently provide — the same gap that leaves the $a(T)$ relation undetermined (NEFOLD-1). *(Scope clarified 05/09/2026: an earlier version stated that the amplitude was "set by the Cartan density". It is not, for the reason above.)*

An important caveat on the normalisation. Converting k_c into comoving units today requires the full expansion history, i.e. the $a(T)$ relation through the stiff era, which this framework does *not* currently determine (item NEFOLD-1). The illustrative normalisation used below, $k_c \simeq 2.2 \times 10^{-4}\, e^{N_{\rm stiff}-57}\,\mathrm{Mpc}^{-1}$, places k_c at today's horizon for $N_{\rm stiff} = 57$; it is a *placeholder*, not a derived anchor. In particular it must not be read as implying that the stiff era supplies the e-folds solving the horizon problem: §2.1 states the opposite — in the stiff era the comoving Hubble radius *grows*, and the ECF *does not* resolve the horizon problem through the stiff era. The topological network is a *candidate* route, not an established one: a post-bounce Kibble–Zurek network cannot by itself correlate regions never in causal contact, and a causal solution would require a pre-bounce/transfer/globality chain that remains OPEN. *The numerical values below therefore indicate the form and the sensitivity of the constraint, not a calibrated threshold.*

With that reservation, requiring $\mathcal{P}_\mathcal{R}(k)$ to stay below observational bounds yields a *lower* bound on the duration of the stiff era:

Probe	k (Mpc^{-1})	$\mathcal{P}_\mathcal{R}$ bound	$N_{\rm stiff}$ min
CMB spectral shape	0.2	2×10^{-10}	58.8
Lyman-α forest	10	10^{-6}	59.9
CMB μ-distortions	10^3	10^{-5}	63.7
Primordial black holes	10^6	10^{-2}	68.3

$$N_{\rm stiff} \gtrsim 68 \qquad \text{(illustrative normalisation)} \tag{30}$$

The *form* of this consequence is falsifiable — a blue component generated by the stiff era necessarily bounds its duration from below — but the *threshold* is not yet calibrated: it inherits the placeholder normalisation discussed above. Establishing $a(T)$ through the stiff era (NEFOLD-1) is what would turn this into a quantitative kill-switch. The estimates above are order-of-magnitude (the $\nu = 0$ mode carries a logarithmic correction, $\mathcal{P}_\mathcal{R} \propto k^3 \ln^2$, not included), and they inherit the uncertainty on k_c discussed next.

***Ab initio* estimate of $N_{\rm stiff}$ (item NEFOLD-1).** During the stiff phase ($w = 1$, $\rho \propto a^{-6}$),

$$N_{\rm stiff} \;=\; \ln\frac{a_{\rm end}}{a_{\rm start}} \;=\; \frac{1}{6}\ln\frac{\rho_{\rm start}}{\rho_{\rm end}}, \tag{31}$$

so $N_{\rm stiff}$ is fixed by the ratio of densities bounding the stiff era. Taking the start at the bounce ($T_{bb} \approx 0.8\,T_{\rm Pl} \sim 10^{19}\,\mathrm{GeV}$), the end at electroweak crystallisation ($T_{\rm EW} \sim 159\,\mathrm{GeV}$, Foundation II) and the sub-dominant radiation scaling $T \propto 1/a$,

$$N_{\rm stiff} \;\approx\; \ln\frac{T_{bb}}{T_{\rm EW}} \;=\; \ln\!\big(6.1\times10^{16}\big) \;\approx\; 39. \tag{32}$$

Honest assessment. This estimate falls *short* of both requirements: of the reference value 57 (the illustrative normalisation that places k_c at today's horizon — not a horizon-problem requirement, since the stiff era does not solve the horizon problem, §2.1), and *a fortiori* of the $\gtrsim 68$ demanded by Eq. (30). The obstacle is precisely identified and is the same in both cases: the result depends critically on the *end* of the stiff phase ($a_{\rm end}$), i.e. on the initial radiation/stiff fraction, which the model does not constrain *a priori*; the choice $T_{\rm end} = T_{\rm EW}$ is motivated but not unique, and $T \propto 1/a$ is only approximate. *Status*: NEFOLD-1 is not resolved. It is now a well-posed problem, and closing it requires deriving the end of the stiff era from reheating thermodynamics — the same missing ingredient that limits the monopole estimate (MONO-1) and the bound of Eq. (30). Deferred to Foundation IV.

3.5 Positioning: ECF vs Hartle–Hawking–Hertog

The no-boundary approach (Hertog–Janssen–Karlsson 2023)

The Hartle, Hawking and Hertog programme [9] is the closest competing approach to the ECF on the problem of the universe's initial conditions. Their fundamental proposal is the *no-boundary wave function* (NBWF): the quantum state of the universe is defined by a Euclidean path integral over compact geometries with no boundary,

$$\Psi[h_{ij}] = \int_{\rm compact} \mathcal{D}g\,\mathcal{D}\phi\; e^{-S_E[g,\phi]}, \tag{33}$$

selecting classical inflationary histories. In a recent work [10], Hertog, Janssen and Karlsson showed that the Kontsevich–Segal (KSW) criterion — applied to the NBWF's complex saddles — acts as a selection mechanism on inflationary potentials, effectively bounding the tensor-to-scalar ratio $r < 0.08$ in the representative models considered, in agreement with Planck.

Convergences with the ECF

Three points of deep convergence:

1. **Abandonment of the initial singularity.** Both approaches replace the Big Bang singularity with a regular quantum state: the NBWF via a compact Euclidean geometry, the ECF via the Poplawski–ECKS torsional bounce. Both satisfy the Wheeler–DeWitt equation.

2. **Initial conditions: partly derived, partly calibrated.** The NBWF fixes the initial conditions through Euclidean compactness. The ECF fixes the initial conditions through κ^2 and $\langle S^2\rangle^{(0)}_{\rm vol}$ (Foundation I). *Qualification added 12/09/2026:* $\kappa^2 = 8\pi G/c^4$ is a constant of the theory, genuinely derived; $\langle S^2\rangle^{(0)}_{\rm vol}$, however, is tied to the condensate amplitude s_0, which is *calibrated* rather than derived. The comparison with the NBWF therefore holds for the structure of the argument, not for the epistemic status of every input: the ECF initial state is *partly* derived and *partly* calibrated.

3. **Selection of classical histories.** The NBWF selects inflationary histories via KSW. The ECF selects the stiff phase via the ECKS Friedmann constraint (§5.4).

Fundamental divergences

Point	Hartle–Hawking–Hertog	ECF (Foundation III)
Initial mechanism	Compact Euclidean geometry	ECKS torsional bounce (Poplawski)
Inflation	Required (KSW-selected)	Replaced by $w = 1$ stiff phase
n_s	$n_s = 0.965$ (slow-roll inflation)	Not predicted — inherited (PO-F3-NS)
n_T	$n_T < 0$ (standard inflation)	$n_T > 0$ **(distinctive prediction)**
Isotropy	Guaranteed by the isotropic NBWF	Open question PO-F3-ISO
Geometric framework	General relativity + scalar field	Einstein–Cartan (non-zero torsion)
Torsion	Absent	Fundamental (source of the bounce)
Kill-switch	$r > 0.08$ — *instrument to be reassessed (DOE/NSF support for CMB-S4 ended (2025))*	$\Pi_{\rm GW} < 0.02$ (LISA, launch planned ∼2035; verdict after data accumulation)

The key observational discriminant: n_T

The most falsifiable divergence between the two approaches is the **sign of the primordial gravitational spectrum**:

- Hartle–Hawking–Hertog: $n_T \leq 0$ (slow-roll inflation, Maldacena relation). The KSW criterion selects $r < 0.08$ but preserves $n_T < 0$.
- ECF: $n_T > 0$ (stiff phase $w = 1$, stochastic enhancement), independent of the n_s tension (PO-F3-NS).

Joint kill-switch (ECF vs HHH): LISA (launch ∼2035) will measure $\Pi_{\rm GW}$ (chiral asymmetry) and n_T (spectral tilt of primordial gravitational waves).

- $n_T > 0$ AND $\Pi_{\rm GW}$ consistent with the calibrated template 0.20 ± 0.05: supports the ECF chiral tensor *sector/template* — not a confirmation of the framework as a whole, since both the blue tilt and $\Pi_{\rm GW}$ depend on the stiff transfer and the chirality channel, which remain conditional (PO-ECF-ACTION). HHH disfavoured.
- $n_T < 0$ AND $r < 0.08$: HHH confirmed, ECF disfavoured.
- $n_T > 0$ AND $\Pi_{\rm GW} < 0.02$: ECF partially disfavoured (chiral sector).

This test is the most powerful current discriminant between the two singularity-free quantum-cosmology programmes.

4 Falsifiable Predictions and Kill-Switches

4.1 ECF observational kill-switches: structured table

Table 2 summarises the falsification criteria, organised by mission and by horizon date. *Scope.* None of these thresholds refutes the framework as a whole: each chain passes through at least

one link that is currently calibrated, phenomenological, conditional or open and indispensable to the observable. Each row therefore tests the sector, realisation or pre-specified template it depends on — which leaves the thresholds themselves sharp and decisive for that sector.

LISA test (launch ∼2035): double discriminant

LISA will simultaneously measure n_T and $\Pi_{\rm GW}$. Three theories compete on these observables. *(Corrected 18/10/2026: a third observable, the post-merger torsional tail $\tau_T \sim 10^6$ s, is withdrawn: at the Macro-Knot radius given by Foundation II, $\tau_T \sim R/c \approx 1$–$1.4$ s, indistinguishable from the general-relativity ringdown.)*

Model	n_T	$\Pi_{\rm GW}$
ECF (Foundation II/III)	> 0	0.20 ± 0.05 (template)
Slow-roll inflation	< 0	≈ 0
Hartle–Hawking–Hertog (2023)	≤ 0	≈ 0

LiteBIRD kill-switch (launch JFY2036): [32, 36] LiteBIRD will measure the CMB polarisation rotation β with a forecast total error budget $\sigma(\beta) \sim 0.02°$ [35] *(corrected 15/10/2026; earlier versions gave 0.1°)*. The ECF predicts $\beta > 0$ (axial torsion breaks parity P by generating a net cosmic birefringence). A non-detection $\beta < 0.1°$ falsifies the ECF's CPT sector independently of LISA.

LISA outcome: if $n_T > 0$ AND $\Pi_{\rm GW}$ is consistent with the calibrated template (0.20 ± 0.05), the ECF chiral tensor sector would be supported; the template being calibrated, not derived, this would not by itself establish the framework. If $n_T \leq 0$, the ECF is falsified independently of $\Pi_{\rm GW}$.

5 Observational Confrontation

5.1 BBN Constraint

The F3-CMB conjecture rests on the assumption that $\rho_{\rm spin}$ is negligible at the epoch of primordial nucleosynthesis ($T_{\rm BBN} \sim 1$ MeV). In the present version this assumption is **not demonstrated**: BBN compatibility of the spin sector is an open question (PO-ECF-SPIN-NORM, PO-ECF-BBN-SPIN).

BBN status: compatibility not demonstrated. Earlier versions of this section presented an analytical estimate, $|\rho_{\rm spin}|/\rho_r \sim 10^{-51}$ at $T_{\rm BBN} = 1$ MeV, as a confirmation with no free parameter. That estimate is withdrawn as a test, for two reasons. (i) It used numerical values of the condensate normalisation ($\langle S^2\rangle^{(0)}_{\rm vol}$ and the resulting $P^{(0)}_{\rm tors}$) that have no producing calculation and are dimensionally inconsistent; these values are withdrawn from Foundation II, and the normalisation remains open (PO-ECF-SPIN-NORM). (ii) It diluted $\langle S^2\rangle_{\rm vol}$ as a^{-6}, whereas Foundation II writes a^{-3} and $a^{-3(1+w)}$ for the same quantity. Whether the stiff spin branch calibrated in Foundation I and the 3D condensate of Foundation II are the same degree of freedom is not settled, the upper limit of validity of the Foundation I branch is not fixed, and no quantitative BBN test of either sector exists in the corpus (PO-ECF-BBN-SPIN). Foundation I [7] now states what that limit must satisfy: extended as a^{-6} to $T \sim 1$ MeV, the calibrated branch gives $\rho_{\rm spin}/\rho_{\rm r} \sim 10^9$–$10^{11}$, which BBN excludes, while the sound-horizon calibration needs the same law up to $z \sim 10^5$; the two Foundation I and II normalisations differ by ~ 22 orders of magnitude. The F3-CMB conjecture therefore carries the BBN assumption as an open condition. The figure of earlier versions (`Fig_ECF_BBN_Constraint.png`) is withdrawn from this document until these two items are addressed.

Observable	**Mission / date**	**ECF confirmation**	**ECF falsification**	**F3 ref.**
$n_T > 0$ (blue GW spectrum)	LISA (∼2035)	$n_T > 0$ detected at $> 2\sigma$	$n_T \leq 0$ confirmed	§5.4
$\Pi_{\rm GW} \simeq 0.20 \pm 0.05$ (GW chirality; calibrated template, chiral sector)	LISA (∼2035)	Chiral asymmetry consistent with 0.20 ± 0.05	$\Pi_{\rm GW} < 0.02$	§3.5
Macro-Knot merger, $f \sim 50\,$mHz (conditional)	LISA (∼2035)	Burst detected	none (tail withdrawn, *18/10/2026*)	§5.3
$f_{\rm NL}^{\rm KZ} > 0$ (sign conditional on the Poisson→ ζ bridge; value open)	future CMB surveys — timescale not fixed	$f_{\rm NL} > 0$ at $> 1\sigma$ ($\sigma \sim 0.3$)	$f_{\rm NL} < -1$ at $> 2\sigma$	§3.1
C_ℓ suppression at $\ell \sim 233$	ACT/SPT and future CMB surveys	$\sim 18\%$ deficit at $\ell \sim 233$	Smooth LCDM spectrum at $\ell = 233$	§3.3
Blue component bounds the stiff era	CMB μ-distortions, PBH	$N_{\rm stiff} \gtrsim 68$ *(threshold not yet calibrated)*	$N_{\rm stiff}$ independently fixed below 68	§3.4
L-amino acids on Mars	MSR — cancelled (2026); no schedule	Biological L-aa confirmed (majority, independent abiogenesis)	*Revised 01/09/2026:* majority D-aa across several independent, confirmed-abiogenic samples — a single D-result no longer suffices alone (see *Foundation IV*, homochirality box: without global network coherence, a minority of D-dominant pockets is expected even if the framework holds; precise probabilistic threshold not yet derived, PO-F3-CHIRAL)	see *Foundation IV*
Micro-Knot aggregates (Poisson $\lambda \geq 36$, $M_{\rm typ} \sim 6.8 \times 10^{-5}\,M_\odot$)	Roman (microlensing)	Excess of events with $t_E \sim$ 1–10 h ($\tau_{\rm micro} \sim 5 \times 10^{-8}$)	No excess in the 1–10 h window ($f_{\rm DM} = 1$ assumed; OGLE-IV: $f_{\rm DM} \lesssim$ 1–2%, PO-F2-10)	Foundation II
Future space tests of the equivalence principle (torsion) — *MICROSCOPE 2 is a mission concept with no fixed schedule*	no schedule	WEP violation: no predicted amplitude (no relation to s_0 established; corrected 22/10/2026)	Conditional test, not a kill-switch (current measurement $\eta(\mathrm{Ti,Pt}) = [-1.5 \pm 2.3_{\rm stat} \pm 1.5_{\rm syst}] \times 10^{-15}$ [39], i.e. $\lvert\eta\rvert \lesssim$ few $\times 10^{-15}$; corrected 22/10/2026)	Foundation I (WEP channel)

Table 2: ECF observational kill-switches, organised by observable, mission, and time horizon. Each row defines a Popper-style falsifiable test: the falsification criterion is precisely quantified, and the anti-escape-hatch is implicit (a detection of opposite sign is not a disguised confirmation). **Priority test:** $n_T > 0$ AND $\Pi_{\rm GW}$ consistent with the calibrated template 0.20 ± 0.05 (LISA, launch planned ∼2035; verdict after data accumulation) discriminates the ECF from both inflation and Hartle–Hawking–Hertog with a single observable. *(Corrected 15/10/2026: the Micro-Knot row, which read "sub-lunar lensing excess / limit $< 10^{-4}\rho_{\rm DM}$" for a single 3.74×10^{24} kg object, is aligned with the Roman entries of Foundation II; Mars Sample Return ("MSR 2028" in earlier versions) is cancelled and MICROSCOPE 2 ("∼2030") has no schedule; cross-references corrected.)*

Distinctive predictions vs inflation

The C_ℓ suppression predicted by the ECF occurs at $\ell \sim \chi_*/\xi_{\rm KZ} \approx 233$ (testable by ACT/SPT and future CMB surveys; *the CMB-S4 sensitivity forecast is no longer available*), *not* at $\ell < 10$. The Planck low-ℓ anomaly remains an open question within the ECF (see PO-F3-ISO, §3.3).

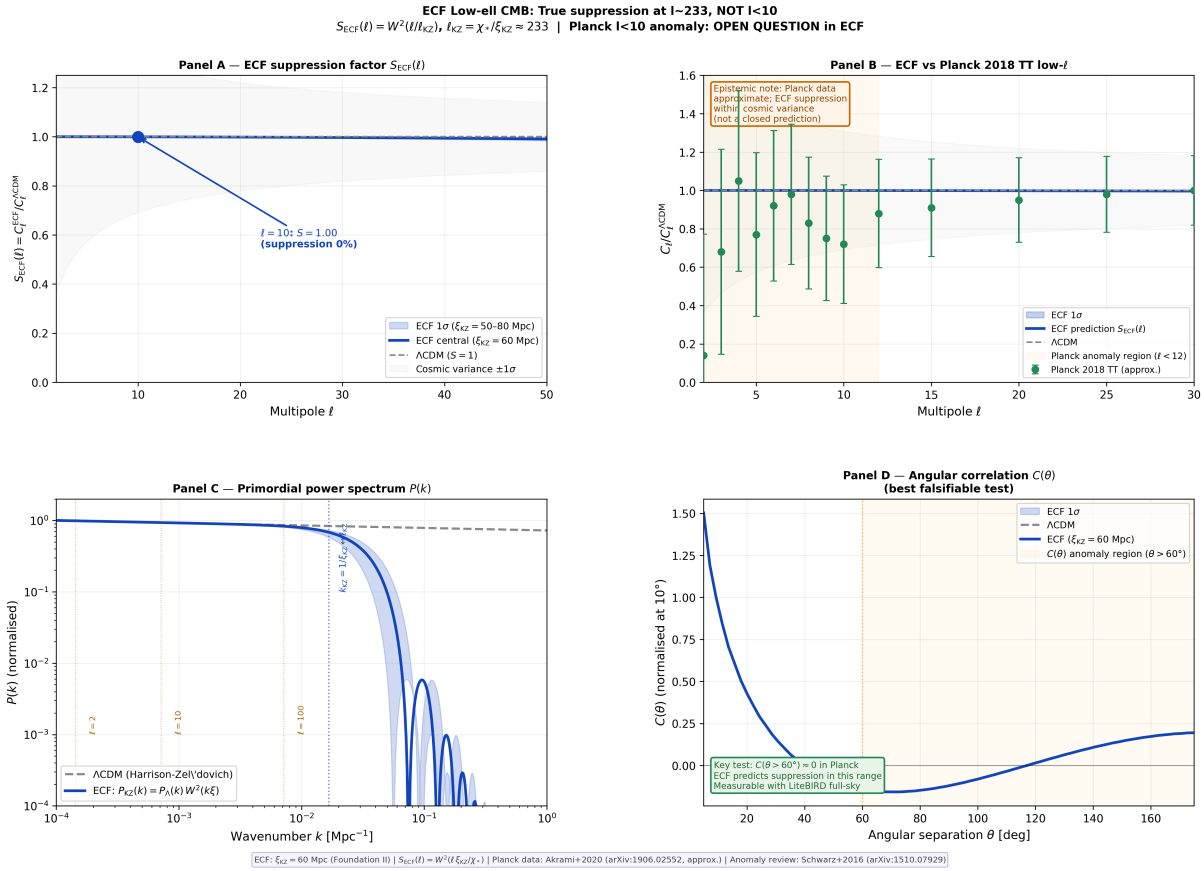


Figure 4: **ECF suppression of the CMB power spectrum: an honest prediction and the open question PO-F3-ISO.** *Panel A:* suppression factor $S_{\rm ECF}(\ell) = W^2(\ell/\ell_{\rm KZ})$ where $\ell_{\rm KZ} = \chi_*/\xi_{\rm KZ} \approx 233$. The suppression is zero at $\ell < 10$ and reaches $\sim 18\%$ at $\ell \sim 233$ (13–31% over the 1σ range of $\xi_{\rm KZ}$, 50–80 Mpc). Blue point: $S(10) = 1.00$ — the ECF predicts *no* suppression at $\ell = 10$. *Panel B:* ECF vs approximate Planck 2018 TT data (Planck 2018 [13]) at low ℓ. The ECF (blue line) is indistinguishable from ΛCDM in this range: the Planck $\ell < 10$ anomaly is not explained. *Panel C:* primordial spectrum $P(k)$ modulated by the KZ shape factor $W^2(k\xi_{\rm KZ})$, $\xi_{\rm KZ}$ being read as the cell radius (the convention of the void-size threshold $R > \xi_{\rm KZ}$); the suppression sets in at $k_{\rm KZ} = 1/\xi_{\rm KZ} \approx 0.017\,{\rm Mpc}^{-1}$, the wavenumber corresponding to $\ell_{\rm KZ}$. *(Corrected 15/10/2026: earlier versions gave $W^2(k\xi_{\rm KZ}/2)$ for this panel, a convention that would place the 18% suppression near $\ell \approx 460$ rather than 233, a cutoff at $2\pi/\xi_{\rm KZ} \approx 0.1\,Mpc^{-1}$, and an 18–22% range over $\xi_{\rm KZ} = 50$–$80\,Mpc$.)* *Panel D:* angular correlation function $C(\theta)$ (the best falsifiable test); the ECF suppression at $\theta > 60°$ is testable with full-sky LiteBIRD. **Epistemic status:** the suppression at $\ell \sim 233$ is a genuine ECF prediction; the Planck $\ell < 10$ anomaly is an open question (PO-F3-ISO, §3.3). *Reproduced by:* `plot_F3_ecf_Cell_suppression.py`.

The idea that the spacetime metric might be an *emergent* phenomenon of quantum vacuum fluctuations — rather than a fundamental postulate — was pioneered by Sakharov [18], who showed that the Einstein-Hilbert gravitational action can be derived as a residual vacuum effect of quantum fields coupled to curvature. This *induced gravity* programme constitutes the direct epistemological precedent of the ECF framework: in our approach, it is torsion — crystallised

from fermionic spin density — that provides the emergent geometric structure [18, 4]. The first models of a non-singular bounce in relativistic gravitational collapse were explored by Novikov [19], who showed that collapse can turn into expansion, laying the groundwork for what Poplawski later formalised within the Einstein-Cartan framework with torsion [4, 5]. The present manuscript, *Foundation III*, develops this programme in the strong-gravity sector: singularity avoidance inside black holes, the transient Einstein-Rosen geometry, and compact-object observables within the ECF framework.

5.2 DESI DR1/DR2 Comparison with Full Covariance

Context and commitment

The PIT [6], Foundation I [7] and Foundation II [8] papers all three explicitly deferred the rigorous comparison of the ECF point $(w_0, w_a) = (-0.904, -0.153)$ — calibrated in Foundation I; $\xi = 1.645$ is a parameter calibrated on the DESI preference used by the ECF, so as to reproduce the target value of w_a; it is therefore not a prediction independent of DESI — with the DESI BAO measurements to Foundation III. Since those papers were written, DESI has published its DR2 results (DESI Collaboration [27]) which strengthen the deviation from ΛCDM to 3.1σ (BAO+CMB; 2.8–4.2σ with SNe), and announced completion of the full 3D map [30]. This is the only commitment present simultaneously in all three papers. This section honours it via the script `Scripts_F1/B_Paper_Plots/plot_ecf_desi_mcmc_v3_0.py` (Foundation I copy, the reference version; GitHub repository [21]).

Method v3.0: full covariance matrix

The previous script version (v2.1) used diagonal errors. Version v3.0 builds the 12×12 block-diagonal covariance matrix published by Adame *et al.* [20]: the 5 intra-bin $(D_M/r_d,\ D_H/r_d)$ pairs are correlated with $\rho \in [-0.389, -0.477]$ (negative correlations, typical of BAO measurements against a coupled dilation background). The likelihood function is:

$$\chi^2_{\rm cov} = \big(\mathbf{y}_{\rm obs} - \mathbf{p}(\boldsymbol{\theta})\big)^\top \mathbf{C}^{-1} \big(\mathbf{y}_{\rm obs} - \mathbf{p}(\boldsymbol{\theta})\big), \tag{34}$$

with $\boldsymbol{\theta} = (\Omega_m, w_0, w_a)$, h fixed by a Gaussian prior (two cases: Planck $h = 0.674$, SH0ES $h = 0.730$). The ECF distance is computed via the Mahalanobis distance in the marginalised (w_0, w_a) space of the MCMC chain:

$$d_{\rm Maha} = \sqrt{\boldsymbol{\delta}^\top \hat{\mathbf{C}}^{-1}_{w_0 w_a} \boldsymbol{\delta}}, \quad \boldsymbol{\delta} = (w_0^{\rm ECF} - \hat{w}_0,\ w_a^{\rm ECF} - \hat{w}_a), \tag{35}$$

where $\hat{\mathbf{C}}_{w_0 w_a}$ is the marginalised covariance matrix estimated from the MCMC chain.

Numerical results

Quantity	Planck prior	SH0ES prior
$w_0^{\rm MCMC}$	-0.45 ± 0.21	-1.02 ± 0.22
$w_a^{\rm MCMC}$	-2.36 ± 1.05	-0.73 ± 1.18
$\rho(w_0, w_a)$	-0.989	-0.989
$\Delta\chi^2$(CPL − ΛCDM)	-7.44	-35.83
ΔAIC(CPL − ΛCDM)	-3.44	-31.83
$\Delta\chi^2$(ECF − ΛCDM)	$+6.21$	$+38.59$
$d_{\rm Maha}$ (ECF vs MCMC, 2D)	$\mathbf{2.03}\,\sigma$	$\mathbf{6.98}\,\sigma$

(Corrected 15/10/2026: $\Delta AIC = \Delta\chi^2 + 2\Delta k$ with $\Delta k = 2$ for CPL; earlier versions gave -11.44 and -39.83, the penalty having been applied with the wrong sign in the producing script.)

Figures

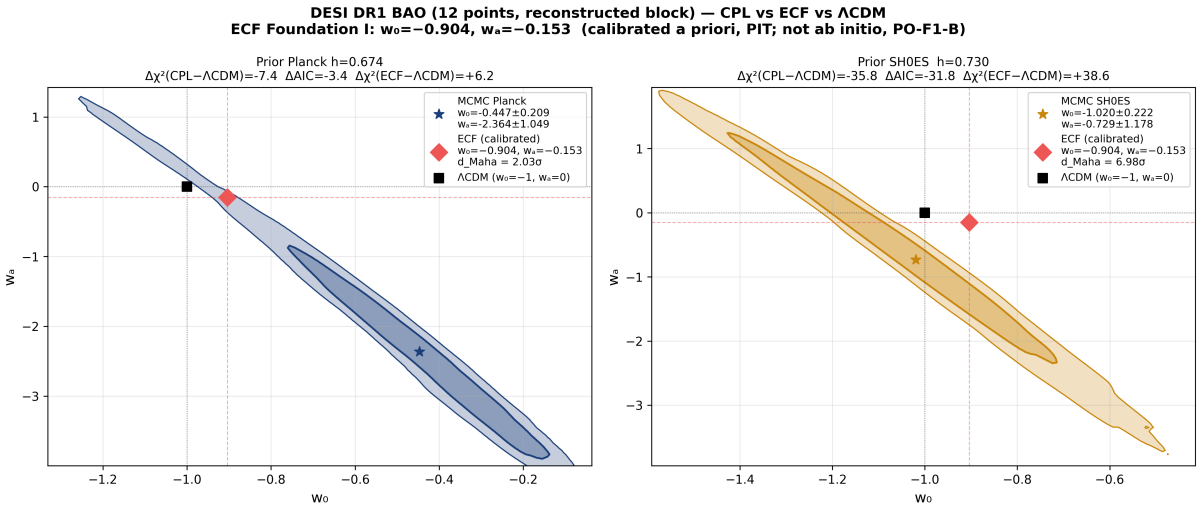


Figure 5: **DESI DR1 BAO MCMC contours (12 observables, reconstructed block) — ECF vs CPL vs ΛCDM.** $1\sigma/2\sigma$ confidence regions marginalised over (w_0, w_a), obtained via Metropolis-Hastings MCMC with the full covariance matrix $\mathbf{C}^{-1}$ of Adame *et al.* [20] (intra-bin D_M-D_H correlations included, $r \in [-0.389, -0.477]$, as printed in Table 1 of that reference). **Left panel:** Planck prior ($h = 0.674$). **Right panel:** SH0ES prior ($h = 0.730$). Symbols: ⋆ MCMC median; ◇ ECF calibrated point $(w_0, w_a) = (-0.904, -0.153)$ (calibrated in Foundation I [7]; w_a through $\xi = 1.645$, calibrated on the DESI preference used by the ECF, hence not independent of DESI; see also [6]); ■ ΛCDM. The Mahalanobis distance is $d_{\text{Maha}} = 2.03\sigma$ (Planck) and 6.98σ (SH0ES) — see §5.2 for interpretation. *Reproduced by:* `Scripts_F1/B_Paper_Plots/plot_ecf_desi_mcmc_v3_0.py`.

Honest interpretation

1. Critical dependence on the H_0 prior. The two cases show radically different CPL best-fits:

- Planck prior ($h = 0.674$): best-fit $\approx (-0.45, -2.36)$ — $w_0 > -1$ today, with a very negative w_a that carries the trajectory into the phantom regime ($w < -1$) at higher redshift, since $w_0 + w_a \approx -2.8$ *(corrected 15/10/2026: earlier versions called $w_0 > -1$ itself "phantom").*
- SH0ES prior ($h = 0.730$): best-fit $\approx (-1.02, -0.73)$ — close to ΛCDM in w_0, but $w_a < 0$.

This dichotomy reflects the unresolved H_0 tension, which destabilises the DESI DR1 (w_0, w_a) plane in the absence of complementary data (full CMB, independent Type Ia supernovae).

2. Correlation $\rho(w_0, w_a) \approx -0.99$. The full covariance reveals a near-perfect degeneracy: the DESI DR1 constraint alone fixes only the combination $w_0 + b\, w_a$ ($b \approx$ const.), leaving an almost free degree of freedom in the perpendicular direction. The calibrated ECF point $(w_0, w_a) = (-0.904, -0.153)$ sits *perpendicular* to this degeneracy (small $|w_a|$, intermediate w_0), which explains the strong sensitivity of d_{Maha} to the H_0 prior.

3. Status of the ECF point. The value $(w_0, w_a) = (-0.904, -0.153)$ is a calibration point (PIT, [6]): $\xi = 1.645$ is a parameter calibrated on the DESI preference used by the ECF, so as to reproduce the target value of w_a; it is therefore not a prediction independent of DESI. It is not derived *ab initio* from the ECKS action: the single-parameter torsion model does not reproduce this pair (cf. Foundation I, open problem PO-F1-B; Foundation II, PO-F2-4).

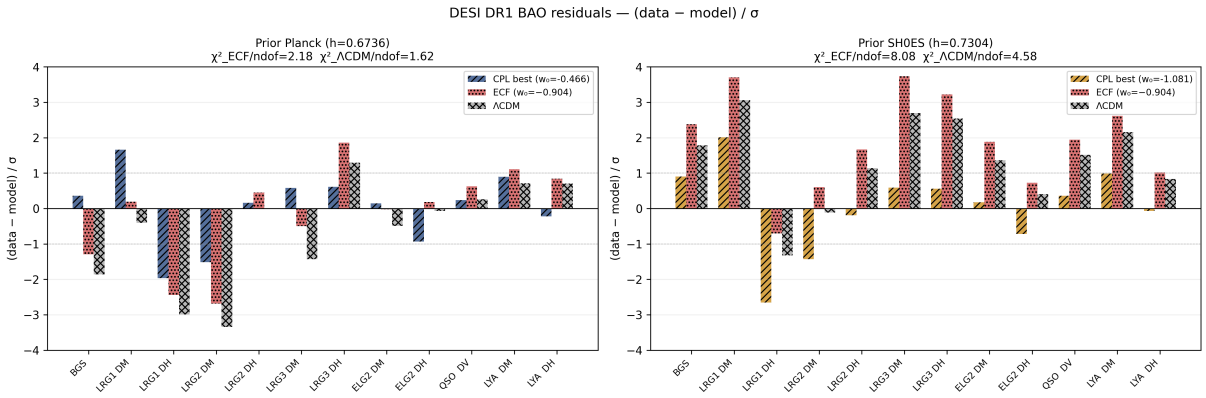


Figure 6: **DESI DR1 BAO residuals by tracer — (data − model)/σ.** Normalised residuals for three models: free CPL best-fit (diagonal hatching), ECF calibrated point $(w_0, w_a) = (-0.904, -0.153)$ (dotted), and ΛCDM (crossed). **Left panel:** Planck prior. The ECF shows systematically shifted residuals on LRG1-2 DH ($\chi^2_{\rm ECF}/\text{ndof} = 2.18$ vs 1.62 for ΛCDM), reflecting the w_a tension under this prior. **Right panel:** SH0ES prior. The ECF residuals are larger ($\chi^2/\text{ndof} = 8.08$), but the SH0ES prior is itself in tension with DESI+CMB (§5.2). *Reproduced by:* `Scripts_F1/B_Paper_Plots/plot_ecf_desi_mcmc_v3_0.py`.

- Under the Planck prior: $d_{\rm Maha} = 2.03\,\sigma$ — *compatible at* 2σ, moderate tension. Note: DESI DR2 [27] indicates $w \neq -1$ at 3.1σ (BAO+CMB), which strengthens the relevance of the CPL comparison but does not modify the pulls computed here (DR1 only; the DR2 analysis with full covariance is deferred to Foundation IV).
- Under the SH0ES prior: $d_{\rm Maha} = 6.98\,\sigma$ — *severe tension*, but the SH0ES prior itself is in $\sim 5\sigma$ tension with DESI+CMB. The SH0ES+DESI DR1 combination is not epistemically coherent without addressing the H_0 tension.

4bis. Putting it in perspective: DES 6-year + DESI DR2 + CMB (May 2026). Since this section was first drafted, the DES collaboration has published its final constraints from six years of data combined with DESI DR2 and CMB (DES Collaboration [26]):

Combination	w_0	w_a
DES 6-year alone	-0.84 ± 0.10	$-0.44^{+0.60}_{-0.55}$
DES 6-year + DESI DR2	$-0.84^{+0.06}_{-0.07}$	$-0.53^{+0.33}_{-0.28}$
DES 6-year + DESI DR2 + CMB	-0.82 ± 0.05	$-0.63^{+0.21}_{-0.18}$
ECF (calibrated)	$\mathbf{-0.904}$	$\mathbf{-0.153}$

The full combination deviates from ΛCDM at 3.0σ, with best-fit parameters *systematically* in the region $w_0 > -1$ and $w_a < 0$.

ECF compatibility (preliminary analysis):

- w_0: $\Delta w_0 = -0.904 - (-0.82) = -0.084$, pull $= -0.084/0.05 \approx -1.7\sigma$ — *compatible.*
- w_a: $\Delta w_a = -0.153 - (-0.63) = +0.477$, pull $= +0.477/0.20 \approx +2.4\sigma$ — *moderate tension.*
- Estimated Mahalanobis distance (without the full DES6yr covariance matrix): $d_{\rm Maha}^{\rm DES6yr} \approx 2.5$–$3.0\,\sigma$.

Honest interpretation: The qualitative direction is *confirmed* — DES6yr and the ECF agree on $w_0 > -1$ and $w_a < 0$ (evolving quintessence). The quantitative tension lies in the amplitude

of w_a: the ECF prediction ($w_a = -0.153$) underestimates the evolution of dark energy compared with the DES6yr observations ($w_a \approx -0.63$). This gap constitutes a significant observational constraint on the ECF dark-energy sector, whose physical origin remains to be identified (see PO-F3-DESI below and Foundation IV).

A degeneracy the background data cannot lift. The comparison above — like every (w_0, w_a) confrontation in this paper — is a *background* comparison, and there is a published limit to what such a comparison can establish. Petri, Marra & von Marttens [28] construct a dark-sector interaction *exactly degenerate with CPL at the background level*, find that both it and CPL fit DESI DR2 + DESY5 + CMB significantly better than ΛCDM, and note that the two are *indistinguishable in background observables*. The agreement reported here therefore constrains the ECF trajectory, but does not by itself establish that the mechanism behind it is torsion rather than a dark-sector interaction. We state this rather than leave it implicit.

Two considerations bear on it. First, Wang, Yu & Wu [29] report that once the equation of state is allowed to evolve — the regime relevant here — the preference for a non-zero dark-sector coupling weakens substantially while the preference for dynamical dark energy persists, the interaction adding essentially nothing to the fit. Second, both works converge on the same discriminant: late-time structure growth, which in the authors' words, contrasting growth signatures *"highlight late-time large-scale-structure observations as a promising avenue for distinguishing dark-energy dynamics from dark-sector interactions"* [29].

That is where the ECF is testable rather than degenerate. The S_8 sector (lowered by the background closure $\Omega_m = \omega_m/h^2$, to $S_8 \simeq 0.780$ in a CAMB ΛCDM reference, the residual correction to 0.766 depending on ECF perturbative dynamics not yet implemented), the scale-dependent $G_{\rm eff}(k)$ (Foundation I) and the void size function of §3.2 are growth-sector predictions. Integrating the Foundation I growth equation on the ECF background (with $\Omega_m = 0.267$) gives an ECF/ΛCDM $f\sigma_8$ ratio that rises from 0.870 to 0.959 across $0 < z < 0.8$ (a 9.8% variation), compared with the sharp low-redshift drop that places the interacting model of [28] in significant tension with growth-rate data (not included in their fit); against a compilation of six published RSD measurements the two histories are statistically indistinguishable ($\Delta\chi^2 = -1.71$ for $N = 6$, with the normalisation $\sigma_8 = S_8/\sqrt{\Omega_m/0.3} = 0.812$, i.e. the target $S_8 = 0.766$ converted at $\Omega_m = 0.267$, not a computed amplitude (a CAMB ΛCDM reference on the same background gives $\sigma_8 = 0.825$); the former normalisation $\sigma_8 = 0.747$ is historical and would give +1.74, `Scripts_F1/compare_fsigma8_data.py`). **This establishes that the ECF growth history is not excluded, not that it is preferred**: with six heterogeneous points, no covariance and no Alcock–Paczynski rescaling, such a $\Delta\chi^2$ carries no evidential weight. A full assessment requires the dedicated perturbative treatment, not undertaken here.

PO-F3-DESI — ECF tension vs multi-probe data (DR1, DR2, DES6yr)

Under the Planck prior, the calibrated ECF point is at $2.03\,\sigma$ from DESI DR1 (full covariance, twelve observables). The DES 6-year + DESI DR2 + CMB combination (DES Collaboration 2026) places the ECF at ≈ 2.5–$3.0\,\sigma$ from the best-fit, mainly via the w_a tension. On the active background ($\Omega_m = \omega_m/h^2 \simeq 0.267$), the χ^2 of the calibrated trajectory on the ten-observable subset of DESI DR1 used in `calibrate_w0wa.py` (diagonal errors; not the global DESI DR1 χ^2) is 55.3, against 13.7 at the former $\Omega_m = 0.315$ (82.9 for $w = -1$ on the same background).

Positive signal: the direction $w_0 > -1$, $w_a < 0$ is confirmed at 3.0σ (DES6yr+DESI DR2+CMB), in qualitative agreement with the ECF.

Open tension: $|w_a^{\rm ECF}| = 0.153$ vs $|w_a^{\rm DES6yr}| = 0.63$ — a gap of a factor ~ 4. This is a marginal comparison, however, and w_0 and w_a are strongly correlated in every fit of this kind; a marginal gap on w_a alone can overstate the true tension. Track: calibrating w_a from the canonical ECF parameters requires integrating the back-reaction effects of torsion on the equation of state, deferred to Foundation IV.

Update (03/09/2026) — the decorrelated pivot comparison. DES Y6 [26] itself reports

a pivot parameter $w_p \equiv w(z_p)$ at the redshift where $\mathrm{Cov}(w_p, w_a) = 0$, which avoids exactly the correlation problem above. Evaluating the calibrated ECF trajectory at that same z_p gives a difference of 0.46σ for DES alone, 0.28σ adding DESI DR2, and 1.45σ for the most constraining combination including CMB — comfortably compatible in every case. This does not resolve the open track above (calibrating w_a from first principles remains deferred to Foundation IV), but it supersedes the 2.5–3.0σ figure quoted from the marginal comparison: on the decorrelated statistic the data actually report, the tension is markedly smaller than the naive (w_0, w_a) gap suggested. *Reproduced by:* `Scripts_F2/compute_ecf_des_pivot.py`.

What is needed to conclude: (i) the full DES6yr covariance for a rigorous Mahalanobis calculation; (ii) resolution of the H_0 tension (Euclid DR1-Complete, mid-2027; $H_0^{\mathrm{DN}} = 73.50 \pm 0.81$ [31] vs Planck 67.4); (iii) computing w_a from the full ECF Lagrangian (Foundation IV). This analysis constitutes the triple PIT+F1+F2 commitment honoured in F3.

4. Impact of the full covariance. The intra-bin correlation $\rho(D_M, D_H) \sim -0.44$ compresses the MCMC contours along the $(w_0 + w_a)$ direction and stretches them in the perpendicular direction. Compared with the diagonal analysis (v2.1): under the SH0ES prior, the diagonal distance was $0.71\,\sigma$ (apparent compatibility) and rises to $6.98\,\sigma$ with the full covariance — illustrating that the diagonal analysis significantly *underestimated* the tension. Under the Planck prior, the diagonal distance was $3.04\,\sigma$ and falls to $2.03\,\sigma$ — the full covariance *improves* compatibility in this case.

5.3 O4 Observational Context: the LIGO Signal S251112cm

O4b context and GWTC-5 (May 2026)

On 26 May 2026, the LVK collaboration published the **GWTC-5.0** [17] catalogue:

- **390 candidates** since 2015, with probability of astrophysical origin $p_{\mathrm{astro}} \geq 0.5$ (161 new from O4b)
- O4b data: April 2024 – January 2025
- First measurement of three ringdown modes of a single black hole (GW250114: two measured, a third constrained), reported in a dedicated LVK analysis [40]; indications of second-generation black holes in GW241011 and GW241110 *(corrected 15/10/2026: "confirmed events" and "harmonics" in earlier versions)*
- New H_0 measurements via standard sirens (GW cosmology)

S251112cm (12 November 2025) was recorded during the third part of O4 (O4c), after the O4b period covered by GWTC-5.0: it is not among the 390 catalogued events, the O4c candidates being left to a later version of the catalogue [17]. Its sub-solar mass (chirp mass between 0.1 and $0.87\,M_\odot$) would set S251112cm apart from all of those events.

Signal description

On 12 November 2025, the LIGO–Virgo–KAGRA (LVK) collaboration reported the candidate S251112cm [11]: a compact object merger with at least one sub-solar-mass component (chirp mass between 0.1 and $0.87\,M_\odot$ [11]; *corrected 15/10/2026: earlier versions gave* $\sim 0.87\,M_\odot$*, the upper end of that range*). Electromagnetic follow-up (DECam, FTW, ZTF) covered 56% of the localisation region with no kilonova counterpart (limit: 92% of KN models excluded by FTW). A Type IIb supernova (SN 2025adtq) was identified in spatial association with $\log_{10}\mathcal{I} \approx 4.8$ [11], i.e. a chance-coincidence probability of $\sim$ 2–9%.

Honest ECF positioning

ECF status of S251112cm: neither confirmation nor falsification

What S251112cm is not, within the ECF: The ECF Micro-Knot mass is $M_{\rm micro} = c^3/(2GH_{\rm EW}) \approx 1.9\times10^{-6}\,M_\odot$, i.e. 5×10^4–5×10^5 times lighter than the S251112cm chirp mass (0.1–0.87 $M_\odot$). The expected GW frequency for a Micro-Knot merger is $f_{\rm GW} \sim 3\,$GHz (Foundation II, peak frequencies of merger signals [8]), outside the LIGO band (10–2000 Hz). *(Corrected 15/10/2026: earlier versions wrote $M_{\rm micro} = c^3/(GH_{\rm EW})$, missing the factor 2 of the canonical horizon mass — the value $1.9\times10^{-6}\,M_\odot$ was already the correct one — and $f_{\rm GW} \sim 10^8$ Hz, an order of magnitude below the merger peak frequency.)* The Macro-Knot ($\sim 10^5\,M_\odot$) is 10^5 times too heavy. **S251112cm is therefore neither a Micro-Knot nor a Macro-Knot ECF candidate.**

What S251112cm contributes to the ECF: (i) A limit on the Micro-Knot density within the LIGO volume: the absence of a signal at $f \sim 3\,$GHz is consistent with the ECF (out of band), without directly constraining it. (ii) The SN IIb + sub-solar GW association: compatible with a Torsion String transit catalysing stellar collapse (Foundation II, §*GMC Transit*), without constituting proof of it. (iii) The discriminating tests of ECF defects are not in the LIGO band: LISA for Macro-Knot mergers at $f \sim 50\,$mHz (*conditional on the existence of the Macro population*; rate not computed, no ECF-specific feature established) and Roman microlensing for Micro-Knots. *(Corrected 18/10/2026: this item named the torsional tail $\tau_T \sim 10^6$ s as the real test; that tail is withdrawn in Foundation II.)*

Methodological lesson

No canonical ECF topological defect has a mass within the LIGO band (0.1–$10^3\,M_\odot$): Micro-Knots ($\sim 1.9\times10^{-6}\,M_\odot$) and Macro-Knots ($\sim 10^5\,M_\odot$) are, respectively, $\sim 5\times10^4$ and 10^2 times out of band. The discriminating test for ECF defects is LISA (Macro-Knot, $f \sim 50\,$mHz) and Roman microlensing (Micro-Knot, $M \sim 3.74\times10^{24}\,$kg).

5.4 Primordial Tensor Spectrum and Amplification During the *Stiff* Phase

Aim and positioning

This subsection does not state a closed prediction for the primordial gravitational spectrum: it clarifies the physical mechanism behind the blue-tilted stochastic background ($n_T > 0$) identified in Foundations I and II [7, 8], identifies the current limits of its derivation, and proposes the perturbative calculation programme that a Foundation IV must carry out. In doing so, Foundation III establishes the link between the bounce dynamics — described at the homogeneous level in Foundation I — and the gravitational observables — structured as kill-switches in Foundation II. The link between the *stiff* phase and the open question of the horizon problem, including the withdrawn topological-network route, is presented in §2.1.

Physical mechanism

Immediately after the non-singular Cartan bounce, the spin density dilutes as

$$\rho_{\rm spin} \propto a^{-6}, \tag{36}$$

generating a transient *stiff* phase with effective equation of state $w = 1$. During this phase, the Hubble rate evolves as $H \propto t^{-1}$ and the effective equation of state exceeds that of radiation ($w > 1/3$). The envelope of sub-horizon modes grows more slowly than in the standard case, so that modes entering the horizon after the bounce retain a higher relative amplitude. The result is a blue-tilted stochastic background ($n_T > 0$), qualitatively distinct from the nearly scale-invariant spectrum associated with minimal slow-roll inflation ($n_T \simeq -r/8 < 0$). This result is analogous to tensor amplification in kination scenarios [15] and constitutes a robust geometric mechanism, independent of the choice of scalar initial conditions.

Expected spectral structure

Subject to the limitations discussed below, the expected structure of the signal can be summarised in three points.

1. **Ultraviolet cutoff.** The non-singular bounce fixes a cutoff scale tied to the minimal scale factor $a_\star \sim 10^{-32}$, beyond which the tensor spectrum is suppressed. In terms of present-day frequencies, this cutoff corresponds to a characteristic frequency

$$f_{\rm cut} \;\sim\; \frac{H_\star}{2\pi}\,\frac{a_\star}{a_0} \;\sim\; 10^{11}\ {\rm Hz}, \tag{37}$$

   where $H_\star$ is the expansion rate at the bounce and a_0 the present-day scale factor. The numerical value in Eq. (37) is an order-of- magnitude estimate; its rigorous derivation requires computing $H_\star$ from the ECKS equations perturbed to first order. With the Cartan density, $H_\star = (8\pi G\rho_c/3)^{1/2} \approx 5\times 10^{43}\,{\rm s}^{-1}$, and $a_\star \sim 10^{-32}$ give $f_{\rm cut} \sim 10^{11}\,{\rm Hz}$, far above the LISA and PTA bands. *Correction (14/10/2026):* earlier versions quoted $\mathcal{O}(1)\,{\rm mHz}$, which does not follow from these inputs.

2. **Spectral rise below $f_{\rm cut}$.** The post-bounce *stiff* phase enhances the high-frequency part of the spectrum, introducing a rising morphology

$$\Omega_{\rm GW}(f)\,h^2 \;\propto\; f^{n_T+2}, \qquad n_T > 0, \tag{38}$$

   up to the break frequency $f_{\rm cut}$. Below the frequency marking the end of the *stiff* era, the spectrum bends and asymptotically joins a plateau or a decline, depending on the phase's effective duration. The exact spectral index n_T depends on the ratio between the duration of the *stiff* phase and the Hubble time, and remains to be computed from first-order tensor perturbation equations.

3. **Chiral polarisation asymmetry.** If the torsional background retains a residual chiral component — encoded in the non-zero vacuum expectation value of the axial torsion $\langle S^\mu\rangle = (s_0, \mathbf{0}) \neq 0$ (see Section 5.4 above, and Foundation II, §*Gravitational Wave Background* [8]) — the amplification may differ slightly between the right- and left-handed tensor sectors, opening the possibility of a net asymmetry $\Pi_{\rm GW} = (P_R - P_L)/(P_R + P_L) \neq 0$. This asymmetry must, however, be treated as a *conditional prediction* of the chiral sector, not as a universal consequence of any Einstein–Cartan cosmology.

Limitations and epistemic status

The preceding statement must be phrased with caution. At this stage, the ECF framework does not yet provide a full tensor transfer function including, in a controlled way: (i) passage through the bounce and the matching of solutions between the contracting and expanding branches; (ii) the full set of reheating, dilution, and particle-production effects in the post-bounce regime; (iii) projection onto the LISA instrument bands (10^{-5}–10^{-1} Hz) and PTAs (10^{-10}–10^{-6} Hz).

The curves and numerical values introduced in Foundation I [7] — $\Omega_{\rm GW}h^2 \sim 10^{-12}$ and $a_\star \sim 10^{-32}$ — and in Foundation II [8] — $h_c(f) \simeq 10^{-23}(f/0.1\,{\rm Hz})^{-1/2}$ and $\Pi_{\rm GW} \simeq 0.20$ — must therefore be understood as *calibrated phenomenological estimates*, useful for identifying an instrumental test zone, but not as the definitive output of a complete tensor cosmological solver. The numerical values and curves presented for $\Omega_{\rm GW}(f)$ must therefore be read as *phenomenological templates* guiding instrument design and analysis strategies, not as predictions from a complete Einstein-Cartan Boltzmann solver. In particular, the detectability of the signal by LISA depends critically on n_T, whose derivation from the ECF bounce geometry is one of the priority open problems of the research programme (cf. *Foundation III — Strong Gravity*, Limitation 1).

Compatibility with current LVK constraints (GWTC-5.0). The most recent LVK analyses [16] detect no statistically significant deviation of gravitational-wave propagation from General Relativity, at the frequencies (10–10^3 Hz) and redshifts ($z \lesssim 1$) currently probed. *Standard-siren* measurements give $H_0 = 71.0^{+9.0}_{-7.1}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (dark sirens and GW170817) [16] *(corrected 22/10/2026: the value* $71.7^{+9.4}_{-7.5}$ *introduced on 15/10/2026 is not in the source)*, compatible with the H_0DN value within error bars. The ECF scenario is fully consistent with these limits: ECF deviations from General Relativity are expected mainly in the high-frequency ($\gtrsim$mHz) primordial stochastic background and in parity-violating tensor modes — two regimes not constrained by current LVK catalogues. LISA (launch ∼2035) and LiteBIRD (launch JFY2036 [36]) are the next instruments capable of testing these predictions. One distinction is worth making explicit, since it explains why the chiral-asymmetry kill-switch of Foundation II ($\Pi_{\rm GW} < 0.02$) is set at LISA rather than at the ground-based network: $\Pi_{\rm GW}$ is a property of the stochastic *background*, whereas the parity tests run on the LVK catalogues — through GWTC-4.0 — measure propagation birefringence of individual events. These are different observables, and no existing analysis constrains the former.

Perturbative calculation programme

The relevant problem breaks down into four ordered steps:

(i) Writing the linearised equations for tensor perturbations $h^{(1)}_{\mu\nu}$ on the effective ECKS background, including the spin-torsion coupling at first order.

(ii) Matching the $h^{(1)}_{\mu\nu}$ solutions across the bounce region, via appropriate junction conditions or a finite-density regularisation.

(iii) Identifying the hinge frequencies associated with the end of the *stiff* regime ($f_{\rm stiff}$) and entry into the radiation era ($f_{\rm rad}$), and computing the index n_T between these two bounds.

(iv) Evaluating the spectral energy densities $\Omega_{\rm GW}(f)$ projected onto the LISA (mHz) and PTA (nHz) bands, accounting for post-bounce entropic dilution.

This programme is formally identified as Limitation 1 of the present manuscript (cf. *Foundation III — Strong Gravity*) and constitutes the immediate horizon for Foundation IV or a standalone article [14]. From this perspective, the phenomenological curves of Foundations I and II must be read as *physical templates* intended to guide the analysis, not as the definitive output of a complete tensor cosmological solver.

6 Conclusion

This document collects the falsifiable cosmological-physics content of the ECF Foundation III programme: the void-statistics kill-switch (a robustness test against DESIVAST), the status of the primordial spectrum (the large-scale tilt is inherited, not predicted, while the stiff era's blue component bounds $N_{\rm stiff} \gtrsim 68$), the BBN question (compatibility not demonstrated), the DESI DR1/DR2 comparison with full covariance — together with the background degeneracy that comparison cannot lift, and the growth-sector observables that can — and the primordial tensor spectrum from the stiff phase. Companion documents extend the programme along three further tracks: the strong-gravity sector (wormhole, Cartan core, CPT cascade), the chiral-bounce sector (Landau–Ginzburg potential, parity breaking), and the exploratory programme (baby universes, fertility curve, astrobiology, Foundation IV). See the corpus scope map for the complete routing.

Open problems retained in this document: PO-F3-ISO (isotropy), PO-F3-NS (spectral index, principal open front), PO-F3-XI ($\xi_{\rm KZ}$ not derived *ab initio*), PO-F3-DESI (tension on

the w_a amplitude), PO-F3-VSF (kill-switch currently out of reach of BGS survey depth). All are documented without occultation.

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